

FLEXIBLE IMPUTATION OF INCOMPLETE NETWORK DATA ^{*}

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Abstract

Sampled network data are common in empirical research because collecting full network information is costly, but using sampled networks can lead to biased estimates. We propose a flexible imputation method for sampled networks and show that downstream empirical analysis using the imputed network yields consistent parameter estimates. Our method separately imputes the part of each missing link that is explained by covariates and the remaining variation not captured by them: the former via projection onto observed covariates, and the latter using local two-way fixed effects. The imputation method avoids parametric assumptions, does not rely on low-rank restrictions, and flexibly incorporates observed covariates and unobserved heterogeneity. We establish entrywise convergence rates for the imputed matrix and prove consistency of GMM estimators based on the imputed network. We further derive the convergence rate of the corresponding estimator in the linear-in-mean peer effects model. Simulations show strong performance of our method both in terms of imputation accuracy and in downstream empirical analysis. We illustrate our method with an application to the microfinance dataset of [Banerjee et al. \(2013\)](#).

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1 Introduction

There is a large and growing literature studying how social networks shape individuals’ behavior and economic outcomes. Examples include how social interactions affect information diffusion ([Banerjee et al. \(2013\)](#), [Beaman and Dillon \(2018\)](#)), test scores ([Sacerdote \(2001\)](#)), and demand for financial assets ([Conley and Udry \(2010\)](#), [Cai, Janvry and Sadoulet \(2015\)](#)). Despite this progress, many empirical conclusions are undermined by the fact that researchers rarely observe the full network. Since collecting full network data is costly, applied work typically relies on sampled networks instead. However, [Chandrasekhar and Lewis \(2011\)](#) demonstrate that using sampled networks in place of the full network may lead to biased estimates of regression coefficients or GMM parameters.

To overcome this limitation, we propose a flexible imputation method for sampled networks and show that downstream empirical analysis using the imputed network delivers consistent parameter estimates. Our method is designed for egocentric sampling, one of the most commonly used designs in empirical work, where researchers randomly sample a set of nodes and elicit all their links, not only within the sample but in the entire network. It separately imputes the part of each missing link that is explained by covariates and the remaining variation not captured by them: the former via projection onto observed covariates, and the latter using local two-way fixed effects.

Our proposed method provides a unified and flexible framework for imputing egocentrically sampled networks generated by dyadic network formation process, accommodating a broad class of network structures and exploiting covariate information, making it particularly well suited to the social networks studied in economics.

First, the proposed method avoids parametric assumptions. Many imputation approaches, such as [Chandrasekhar and Lewis \(2011\)](#), [Breza et al. \(2020\)](#) and [Boucher and Houndetoungan \(2020\)](#), rely on knowing the functional form of the network formation model. In practice, however, applied researchers rarely possess such prior information. By contrast, our approach is fully nonparametric and therefore more robust to model misspecification.

Second, our imputation method does not rely on low-rank assumptions. In the statistical analysis of networks, it is common to assume that the underlying network formation model is low-rank and to exploit this structure for imputation; see, for example, [Li et al. \(2023\)](#). However, this assumption fails when the network formation process is nonlinear. For instance, the model in [Graham \(2017\)](#), which is widely used in empirical economics, generally does not exhibit a low-rank structure. Instead, building on ideas from [Freeman and Weidner \(2023\)](#), and [Feng \(2023\)](#), we approximate the nonlinear graphon using locally two-way fixed effects model, without imposing global low-rank structure.

Third, our method leverages observed covariates and accommodates unobserved heterogeneity in link formation. Social networks typically exhibit homophily, where individuals with similar characteristics are more likely to form links, as well as degree heterogeneity, where some individuals tend to have more connections than others. Therefore, an effective imputation procedure should take advantage of available covariate information while also accommodating unobserved heterogeneity in link formation. Our approach achieves both objectives without imposing assumptions on the joint distribution of latent factors and observed covariates.

Our general setting for the network formation model leads to a technical challenge: the latent factors cannot be identified when the network formation model is unknown and potentially nonlinear. As a result, we cannot first estimate the latent factors, as is commonly done in linear factor models, and then estimate the formation process to impute the missing links. To address this difficulty, we build on the idea, first developed by [Zhang, Levina and Zhu \(2017\)](#), that similarity in observed connections can reveal similarity in latent factors. Using the observed subnetwork, we construct a pseudo-distance that serves as a proxy for latent distance. For any unsampled individual, we then find sampled individuals with

small pseudo-distances and use their observed connections to impute that individual’s missing links.

Our main theoretical result for imputation is Theorem 2, which, as the network size and the number of sampled nodes go to infinity, establishes the sup-norm convergence rate of the proposed imputation and provides uniform asymptotic orders for the bias and variance of the imputation error. Such a uniform result is particularly useful when empirical analysis require accurate predictions of link-level conditional probabilities. Because our method relies on a pseudo-distance that approximates latent similarity, the imputation inevitably introduces approximation error. Theorem 2 shows that this error has an asymmetric impact on the imputation error: the leading asymptotic order of the bias does not depend on the pseudo-distance error, whereas the leading order of the variance may be affected by it.

We further show in Theorem 3 that, under suitable regularity conditions, replacing the sampled network with our imputed network yields consistent estimates of the GMM parameters. This result is particularly relevant in empirical applications, where interest typically centers on the consistency of parameter estimates constructed from the imputed network rather than on the accuracy of the imputed links themselves.

Characterizing how imputation error affects the estimation of model parameters is technically challenging. The difficulty arises not only because the imputation error for each link is nonclassical, but also because empirical analyses typically aggregate network data into network statistics, which leads to aggregated nonclassical measurement error and is more complex analytically. To examine how imputation errors influence parameter estimation, we analyze the linear-in-mean peer effects model and derive the convergence rate of the estimator based on the imputed network. Theorem 4 implies that the bandwidth that minimizes the mean-squared error of link-level imputation has a different asymptotic order from the bandwidth that minimizes the mean-squared error of the estimator in empirical analysis. In particular, the bias component of the link-level imputation error plays a more prominent role in determining the estimator’s convergence rate.

We provide extensive simulation evidence to evaluate the performance of our imputation method. First, we compare its imputation accuracy with alternative approaches and show that our method performs well in finite samples across a wide range of sampling rates and network sparsity levels. We then evaluate the performance of our method in empirical analysis, including regressions based on network statistics and the linear-in-mean peer effects model. These simulations demonstrate that the estimates of regression coefficients and GMM parameters based on our imputed networks maintain strong finite-sample performance. Finally, we illustrate the practical usefulness of the proposed method through an empirical application to the study of how social networks affect microfinance adoption (Banerjee et al. (2013)).

Related literature We contribute to the literature on estimating model parameters with sampled network data by developing an imputation method that allows for a fully nonparametric network formation model. Chandrasekhar and Lewis (2011) show that models estimated using sampled networks are generally biased. They propose an analytical correction for certain special cases, and for more general settings, they estimate a parametric network formation model and conduct GMM using simulated moments. Boucher and Houndetoungan (2020) study the estimation of peer effects through social networks when the network data are corrupted, taking egocentrically sampled networks as a special case. Their approach relies on obtaining a consistent estimator of the distribution of the network, which is typically available only under parametric specifications. Herstad (2023) also examine peer-effects estimation with missing links, focusing on settings with large networks and relying on the network formation model of Graham (2017). A key distinction from this line of work is that our method does not rely on parametric specifications of the network formation process and imposes no restrictions on the form of unobserved heterogeneity. Therefore, both imputation and model estimation based on our imputed network are

robust to misspecification.

Additional studies address closely related questions. [Breza et al. \(2020\)](#) demonstrate that aggregated relational data can help recovering network structure without observing any links. [de Paula, Rasul and Souza \(2025\)](#) study how network ties can be identified from panel data. [Thirkettle \(2019\)](#) show that network statistics are set identified under partially sampled networks, although their analysis relies on a strategic network formation model, whereas ours is based on a dyadic formation framework. Other sampling designs have also been studied, for example, induced graph in [Chatterjee \(2015\)](#) and censored network in [Griffith \(2022\)](#).

Our work also relates to the growing literature on estimation with mismeasured network data, as empirical analysis based on imputed networks inevitably inherits the errors introduced in the imputation stage. For peer-effects models, [Lewbel, Qu and Tang \(2024\)](#) show that when the number of misclassified links grows at a rate slower than the network size, the 2SLS estimator of [Bramoullé, Djebbari and Fortin \(2009\)](#) is consistent and standard inference methods remain valid. [Lewbel, Qu and Tang \(2025\)](#) propose an adjusted 2SLS estimator for peer-effects models under i.i.d. link misclassification. Misclassification: [Hardy et al. \(2024\)](#) also study link misclassification under an i.i.d. error structure and obtain consistent treatment effects using an expectation maximization algorithm. [Cai \(2022\)](#) examine linear regression on centrality measures under network sparsity and link-level i.i.d. Bernoulli errors, and propose bias-correction and inference methods for OLS estimators. The setting we consider is technically more challenging than current literature. The imputed network in our framework contains errors that are dependent across links and may have bias of the same order as their standard deviation. Our results characterize the convergence rate of peer-effects estimators in settings that allow for far more general forms of network measurement error.

Methodologically, our approach is related to causal inference using panel data with staggered adoption, as the two settings share similar missing-data structures and face similar challenges associated with unobserved heterogeneity. This line of work includes the rapidly growing literature on synthetic control and causal matrix completion, such as [Arkhangelsky et al. \(2021\)](#), [Athey et al. \(2021\)](#), [Bai and Ng \(2021\)](#), [Agarwal et al. \(2023\)](#), among others. Most of these methods assume that the potential outcomes follow a low-rank structure plus idiosyncratic noise. The work that is most directly connected to ours is the recent literature on nonlinear factor models in panel-data causal inference, particularly the methods of [Feng \(2020\)](#), [Feng \(2023\)](#), and [Deaner, Hsiang and Zeleneev \(2025\)](#), which also allow for nonparametric and nonlinear factor structures. Our local TWFE imputation is analogous to a local linear regression and therefore achieves a second-order approximation error. In contrast, the estimator in [Deaner, Hsiang and Zeleneev \(2025\)](#) corresponds to a local constant regression and attains only a first-order approximation error. The methods of [Feng \(2020\)](#) and [Feng \(2023\)](#) can be viewed as local polynomial regressions that fully exploit smoothness of nonlinear functions. Our approach differs in that it does not require specifying the latent factor dimension and is less sensitive to misspecification of smoothness conditions. An additional advantage of our method is that it leverages observed covariates in a flexible manner to impute missing values.

Finally, our analysis is connected to several strands of the econometrics and statistics literature. The use of pseudo-distance as a measure of latent similarity was first developed by [Zhang, Levina and Zhu \(2017\)](#) and has since been applied in a range of contexts, including nonparametric graphon estimation ([Zhang, Levina and Zhu, 2017](#); [Zeleneev, 2020](#)), controlling for unobservables using network data ([Auerbach, 2022](#)), and nonlinear factor models ([Feng, 2020, 2023](#); [Beyhum and Mugnier, 2024](#); [Mugnier, 2025](#); [Deaner, Hsiang and Zeleneev, 2025](#)). Our analysis is also related to the graphon estimation literature ([Gao, Lu and Zhou \(2015\)](#), [Xu \(2018\)](#)), and dyadic network formation literature ([Graham \(2017\)](#), [Dzemski \(2019\)](#), [Chen, Fernández-Val and Weidner \(2021\)](#), [Ma, Su and Zhang \(2022\)](#)). Finally, extracting the component of the network explained by covariates relies on the dyadic nonparametric regression

developed by [Graham, Niu and Powell \(2021\)](#).

The rest of the paper is organized as follows. In Section 2 we introduce the framework and estimation procedure. Section 3 provides statistical guarantees for imputation. Section 4 gives theoretical properties for the estimates in GMM parameters. Section 5 presents Monte Carlo results for both imputation and downstream empirical analysis. Empirical application to [Banerjee et al. \(2013\)](#) is given in Section 6. Section 7 concludes.

2 Method

2.1 Network Formation Model

Let $A \in \{0, 1\}^{N \times N}$ denote the adjacency matrix of an unweighted network, where $A_{ij} = 1$ if there is a link between i and j and $A_{ij} = 0$ otherwise. We impose $A_{ij} = A_{ji}$ for all i, j and $A_{ii} = 0$, so that the network is symmetric and has no self-loops. For each individual i , let $X_i \in \mathbb{R}^{d_x}$ denote the observed covariates (e.g., gender, education, and other demographic characteristics), and let $\xi_i \in \mathbb{R}^{d_\xi}$ denote the latent factors. We write $X = \{X_i \mid i = 1, 2, \dots, N\}$ and $\xi = \{\xi_i \mid i = 1, 2, \dots, N\}$ for brevity. The network A is generated according to the following process. Conditional on (X, ξ) , the links $\{A_{ij}\}_{1 \leq i < j \leq N}$ are generated independently with probability $\mathbb{P}(A_{ij} = 1 \mid X, \xi) = f(X_i, \xi_i, X_j, \xi_j)$, where $f : \mathbb{R}^{d_x + d_\xi} \times \mathbb{R}^{d_x + d_\xi} \rightarrow [0, 1]$ is symmetric function, referred to as the graphon, whose functional form is unknown to econometricians. Then it is convenient to write A_{ij}

$$A_{ij} = f(X_i, \xi_i, X_j, \xi_j) + \epsilon_{ij}, \quad \forall 1 \leq i < j \leq N \quad (2.1)$$

where ϵ_{ij} are realization error, which is conditionally independent across pairs $\{(i, j) \mid 1 \leq i < j \leq N\}$ given (X, ξ) .

The network formation model (2.1) is a conditionally independent dyadic model, because the graphon f depends only on the characteristics of the pair (i, j) and not on those of other individuals or links in the network. While A_{ij} and A_{ik} ($k \neq j$) may exhibit correlation unconditionally, they are independent given (X, ξ) . This setting is flexible enough to cover a wide range of network formation models commonly used in economics and statistics. Below, we provide several examples.

Example 1 (Stochastic block model) *A widely used example consistent with our network formation framework is the stochastic block model (SBM). It assumes that each individual belongs to one of G groups, $g \in \{1, \dots, G\}$, and that the link formation probability depends only on group membership, that is, the probability of a link between two individuals is determined solely by the groups to which they belong, so that individuals within the same group have the same connection probabilities toward other individuals. The stochastic block model corresponds to a special case of our network formation process where the support of (X_i, ξ_i) is finite. Specifically, if (X_i, ξ_i) can take only G distinct values, the graphon g reduces to a piecewise constant function defined over G^2 group pairs.*

Example 2 (Network formation with transferable utility) *Suppose that individuals i and j form a link if the total surplus from doing so is positive:*

$$A_{ij} = \mathbf{1}(\omega(X_i, X_j)' \beta + g(\xi_i, \xi_j) - U_{ij} \geq 0), \quad (2.2)$$

where the total surplus consists of three components:

- **(Homophily in observables)** $\omega(X_i, X_j)$ measures the distance between observable characteristics of i and j . For instance, one may define $\omega(X_i, X_j) = (|X_{i1} - X_{j1}|, \dots, |X_{id_x} - X_{jd_x}|)'$ for absolute

differences, or $\omega(X_i, X_j) = ((X_{i1} - X_{j1})^2, \dots, (X_{id_X} - X_{jd_X})^2)'$ for squared differences. A negative coefficient $\beta < 0$ captures homophily, meaning that individuals with more similar observables are more likely to form a link.

- **(Individual heterogeneity)** The function $g(\xi_i, \xi_j)$ captures the influence of individual heterogeneity. For example, one may specify $g(\xi_i, \xi_j) = \xi_{i1} + \xi_{j1} - (\xi_{i2} - \xi_{j2})^2$, where the additive component $\xi_{i1} + \xi_{j1}$ induces degree heterogeneity (higher ξ_i implies more connections), while $-(\xi_{i2} - \xi_{j2})^2$ captures homophily in unobservables.
- **(Idiosyncratic shocks)** $\{U_{ij}, 1 \leq i < j \leq N\}$ captures the preference shock that is independent of (X, ξ) . Let $F(\cdot)$ be the cumulative distribution of U_{ij} , then $\mathbb{P}(A_{ij} = 1 \mid X, \xi) = F(\omega(X_i, X_j)' \beta + \xi_i + \xi_j)$. Specifically, when U_{ij} follows type-I extreme value distribution, we have

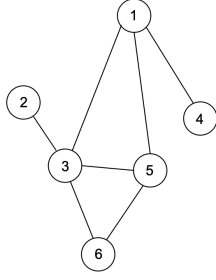
$$f(X_i, \xi_i, X_j, \xi_j) = \frac{\exp(\omega(X_i, X_j)' \beta + g(\xi_i, \xi_j))}{1 + \exp(\omega(X_i, X_j)' \beta + g(\xi_i, \xi_j))}$$

When $g(\xi_i, \xi_j)$ includes only the additive component $\xi_i + \xi_j$, the model reduces to the model studied in [Graham \(2017\)](#), which has been widely used in empirical applications.

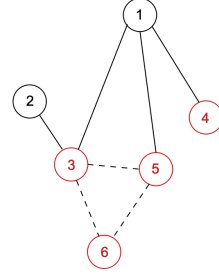
Beyond these examples, our network formation framework can flexibly incorporate richer forms of heterogeneity. For example, the homophily effect on observables, β in (2.2), may depend on the latent factors (ξ_i, ξ_j) , generating heterogeneous slope coefficients. Similarly, the unobserved heterogeneity term $g(\xi_i, \xi_j)$ can represent more complex interaction structures, such as interactive fixed effects $\xi_i' \xi_j$. However, the dyadic nature of our network formation model excludes strategic network formation, in which individuals' decisions to form links depend on that of the others, see for [de Paula \(2020\)](#).

2.2 Data

Economists observe individual covariates X_i (e.g., gender, age) for all i in the network but do not observe the full network A ; instead, only a sampled network is available. In this paper, we focus on a sampling method, egocentric sampling which is widely used in empirical work. Specifically, we randomly sample $n < N$ nodes in the network and ask all their friends not only in the sample but in the whole network. Let $\mathcal{S} \subset \{1, 2, \dots, N\}$ denote the sets of sampled agent with $|\mathcal{S}| = n$. Then a link A_{ij} is observed if $i \in \mathcal{S}$ or $j \in \mathcal{S}$, whereas links between two non-sampled nodes (i.e., $i \notin \mathcal{S}$ and $j \notin \mathcal{S}$) are unobserved. A simple example follows to illustrate the setup.



(a) Full network



(b) Sampled network by sampling $\{1, 2\}$, red nodes are not in the sample, dashed edges are not observed

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ \hline 1 & 1 & \times & \times & \times & \times \\ 1 & 0 & \times & \times & \times & \times \\ 1 & 0 & \times & \times & \times & \times \\ 0 & 0 & \times & \times & \times & \times \end{pmatrix}$$

(c) From full adjacency matrix to partially observed adjacency matrix, $\times$ denotes missing entry

Figure 1: An example of how egocentric sampling works

Since it is no harm to relabel the agents, we can do so such that $|\mathcal{S}| = \{1, 2, \dots, n\}$, in which case the observed adjacency matrix A^{obs} admits a block structure:

$$A^{\text{obs}} = \begin{pmatrix} A_{\mathcal{S}\mathcal{S}} & A_{\mathcal{S}\mathcal{S}^c} \\ A_{\mathcal{S}^c\mathcal{S}} & \textcolor{red}{A_{\mathcal{S}^c\mathcal{S}^c}} \end{pmatrix} \quad (2.3)$$

where $A_{\mathcal{S}\mathcal{S}}$ consists of links between two sampled individuals, $A_{\mathcal{S}^c\mathcal{S}} = A'_{\mathcal{S}\mathcal{S}^c}$ contains links between one sampled and one non-sampled agent, and the bottom right block $A_{\mathcal{S}^c\mathcal{S}^c}$ is unobserved because both agents are non-sampled; we highlight this block in red.

2.3 Imputation

To illustrate the main idea of our imputation approach, let us start with a simple case in which both X_i and ξ_i are observable for every individual in the network. In this case, estimating f_{ij} would be straightforward: we could estimate f_{ij} nonparametrically from the observed adjacency matrix to obtain estimates $\hat{f}_{ij}$, and then use these estimates to impute the missing links in the submatrix $A_{\mathcal{S}^c\mathcal{S}^c}$.

The key difficulty, however, is that the latent factors ξ are unobserved for all nodes. To clarify the main challenge, let us first abstract from the observed covariates X and focus on the case where link formation depends solely on the latent factors. In this setting, the imputation problem reduces to recovering information about ξ from the observed subnetwork in order to predict the missing part of the adjacency matrix. Our approach is motivated by the following idea. Under the network formation model, when two individuals have similar ξ s, their corresponding rows in the (denoised) adjacency matrix are expected to be similar. This observation provides a useful heuristic: similarity in observed connections can indicate similarity in latent positions. Based on this idea, for any individual i who is not in the

sample, we find a set of sampled individuals whose latent factors are likely to be close to ξ_i by comparing their connection patterns in the observed subnetwork. We then use the observed connections of these sampled individuals, weighted by their similarity to i , to impute the missing links for i .

Since the recovered information about ξ from the network is noisy whereas X is accurately measured, it is inappropriate, in the presence of X , to ignore the observed covariates or absorb them into the latent factors, as doing so would discard accurately measured information contained in X . When X is available, our method first extracts the component of link formation explained by the observed covariates and then applies the latent-factor-based procedure described above to the residuals not captured by X . We then combine these two components to obtain the final imputation of the network.

2.3.1 Extract information explained by X

When individual covariates X are available, it is natural to first extract from A the component that can be explained by X . Consider the decomposition

$$f_{ij} = \underbrace{\mathbb{E}(f_{ij} \mid X_i, X_j)}_{\Pi_{ij}} + \underbrace{f_{ij} - \mathbb{E}(f_{ij} \mid X_i, X_j)}_{\Pi_{ij}^\perp}$$

where Π_{ij} represents the part of the graphon that can be explained by the observed covariates, and $\Pi_{ij}^\perp$ captures the residual component of the graphon that cannot be explained solely by the observed covariates. Note that the observed link A_{ij} can then be written as: $A_{ij} = \Pi_{ij} + (\Pi_{ij}^\perp + \epsilon_{ij})$, and the composite error term satisfies $\mathbb{E}(\Pi_{ij}^\perp + \epsilon_{ij} \mid X_i, X_j) = 0$, which implies that a nonparametric regression of A_{ij} on (X_i, X_j) consistently estimates Π_{ij} . [Graham, Niu and Powell \(2021\)](#) derives the pointwise and uniform minimax risks for estimating such conditional expectations and shows that the Nadaraya-Watson (NW) estimator attains the optimal convergence rate when higher-order kernels are used. In practice, however, we use local polynomial regression rather than higher-order kernels, as it is more commonly used. In addition, this nonparametric regression can be implemented using subset of A since A is partially observed. The implementation details are discussed later.

Notably, because the covariates X are observed for every individual in the network, the nonparametric estimator $\hat{\Pi}_{ij}$ can be computed even when both i and j are unsampled. In the next two steps, we focus on estimating this residual component $\Pi^\perp$ using pseudo distance and two-way fixed-effect imputation procedure described below.

2.3.2 Calculate similarity

In this step, we describe how to infer the similarity between latent factors using the observed network links.

For notational simplicity, let $\zeta_i := (X_i, \xi_i)$ for each i , and $f_{ij} := f(\zeta_i, \zeta_j)$ for all $1 \leq i, j \leq N$. Collect the elements f_{ij} into a matrix $F \in \mathbb{R}^{N \times N}$, and the elements ϵ_{ij} into a matrix $E \in \mathbb{R}^{N \times N}$. Then the adjacency matrix A can be expressed as $A = F + E$.¹ Our approach relies on the following idea. For any unsampled individual i and sampled individual j , if their denoised rows in observed network (or equivalently, the corresponding rows in F) appear sufficiently similar, then under suitable regularity conditions, we can say that ζ_i and ζ_j are also close in the latent space.

¹Strictly speaking, the representation above is slightly imprecise. Because self-loops are ruled out, the diagonal entries of F are set to zero rather than f_{ii} . Likewise, we impose $\epsilon_{ii} = 0$ for each i .

Following [Zhang, Levina and Zhu \(2017\)](#), we define the pseudo-max distance between i and i' :

$$\hat{d}_{ii'} = \max_{k=1,\dots,n} \left| \frac{1}{n} \sum_{\ell=1}^n A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|, \quad i, i' = 1, 2, \dots, N$$

The definition above relies only on the friendships of sampled individuals and can therefore be computed from the observed subnetwork. We will show that the pseudo-distance $\hat{d}$ effectively de-noises the adjacency matrix A , so that the distances computed from A and from the noise-free matrix F are asymptotically equivalent. In addition, under regularity conditions discussed later, a small value of $\hat{d}_{ii'}$ implies that the corresponding latent characteristics ζ_i and $\zeta_{i'}$ are sufficiently close in the latent space.

It is important to note that the discussion here concerns the composite characteristic ζ_i , rather than the observable covariate X_i alone. This is because (i) X_i and ξ_i enter the network formation process in a non-separable way, making it impossible to isolate their effects without additional assumptions; and (ii) the information in X_i that can be directly explained has already been extracted in the first step, so there is no need to distinguish between X_i and ξ_i in this step.

2.3.3 Two-way fixed effect imputation

We employ the distance measure $\hat{d}$ obtained in the previous step to construct a local estimator for $\Pi_{ij}^\perp$, referred to as the two-way fixed-effect imputation. Specifically, for an unsampled individual i and sampled individual j , we first estimate the local fixed effects $\hat{\alpha}_i$ and $\hat{\alpha}_j$ by solving

$$\hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{n+2}}{\operatorname{argmin}} \sum_{\substack{i' \in \mathcal{S} \cup \{i\}, j' \in \mathcal{S} \cup \{j\} \\ (i', j') \neq (i, j)}} \left(A_{i'j'} - \hat{\Pi}_{i'j'} - \alpha_{i'} - \alpha_{j'} \right)^2 K_h(\hat{d}_{ii'}) K_h(\hat{d}_{jj'}) \quad (2.4)$$

where $K_h(\cdot)$ is a kernel function with bandwidth h , and $\hat{d}$ denotes the max-pseudo distance. Then the imputed $\Pi_{ij}^\perp$ is given by $\hat{\Pi}_{ij}^\perp = \hat{\alpha}_i + \hat{\alpha}_j$, and the imputed network link is given by:

$$\hat{A}_{ij} = \hat{\Pi}_{ij} + \hat{\Pi}_{ij}^\perp = \hat{\Pi}_{ij} + \hat{\alpha}_i + \hat{\alpha}_j$$

The imputation is “local” in the sense that, by using a kernel estimator, when estimating $\Pi_{ij}^\perp$, we rely only on information from individuals i' and j' whose max-pseudo distances $\hat{d}_{ii'}$ and $\hat{d}_{jj'}$ (or, equivalently, the distances between their latent positions) are sufficiently small. Note that, although the notation omits the indices (i, j) for simplicity, the estimator $\hat{\alpha}$ in (2.4) is pair-specific and thus needs to be re-estimated for each (i, j) .

The procedure is referred to as a “two-way fixed-effect” method because (2.4) takes the form of a two-way fixed-effect model rather than a local constant estimator. This is analogous to replacing a local constant estimator with a local polynomial estimator in nonparametric regression: incorporating the two-way fixed effects provides a higher-order local approximation and thus improves estimation accuracy. We will formally justify this result later using Taylor expansion.

Based on the above steps, we can impute all missing links. By combining the imputed entries with the observed entries of the adjacency matrix, we obtain the imputed adjacency matrix $\hat{A}$. To facilitate understanding of the main idea of our algorithm, we summarize below a simplified version of our method that excludes cross-fitting and cross-validation.

Algorithm 1 Imputation (Basic Idea)

Input: Kernel $K(\cdot)$, bandwidths h , incomplete network A^{obs} , covariate matrix X .

Output: Imputed network $\hat{A}$.

Step 1: Extract information explained by X

Run dyadic nonparametric regression to obtain $\hat{\Pi}_X(\cdot, \cdot)$.

Step 2: Calculate similarity

For each $i \in \mathcal{S}^c$ and $i' \in \mathcal{S}$, calculate the pseudo distance using observed adjacency matrix:

$$\hat{d}_{ii'} = \max_{k=1, \dots, n} \left| \frac{1}{n} \sum_{\ell=1}^n A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|$$

Step 3: Local imputation

For each missing entry (i, j) , i.e., $i \in \mathcal{S}^c$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{n+2}}{\text{argmin}} \sum_{\substack{i' \in \mathcal{S} \cup \{i\}, j' \in \mathcal{S} \cup \{j\}, \\ (i', j') \neq (i, j)}} \left(A_{i'j'} - \hat{\Pi}_{i'j'} - \alpha_{i'} - \alpha_{j'} \right)^2 K_h(\hat{d}_{ii'}) K_h(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij} = \hat{\alpha}_i + \hat{\alpha}_j + \hat{\Pi}_{X,ij}$. By combining the imputed entries with the observed entries of the adjacency matrix, we obtain the (preliminary) imputed adjacency matrix $\tilde{A}$. Truncate $\tilde{A}_{ij}$ to get

$$\hat{A}_{ij} = \begin{cases} \tilde{A}_{ij}, & \text{if } \tilde{A}_{ij} \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij} > 1 \\ 0 & \text{if } \tilde{A}_{ij} < 0 \end{cases}$$

and zero the diagonal of $\hat{A}$ such that $\hat{A}_{ii} = 0$ for all i .

2.3.4 Cross-validation and sample-splitting

Although the proposed algorithm is straightforward to implement, deriving the asymptotic properties of $\hat{A}_{ij}$ can be technically challenging because of the potential dependence between the estimated max-pseudo distance $\hat{d}$ and the error terms ϵ in the two-way fixed-effect estimation.

A standard approach to eliminate this dependence is to employ cross-fitting, in which the data are partitioned into folds so that the first-step and second-step estimations are conducted on separate subsets of the data. Specifically, we split the sample $\mathcal{S}$ into two parts, $\mathcal{S}_1$ and $\mathcal{S}_2$, of equal sizes. We then use only the friendships of $\mathcal{S}_1$ to calculate the similarity $\hat{d}(\cdot, \cdot)$:

$$\hat{d}_{ii'} = \max_{k \in \mathcal{S}_1} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|, \quad i \in \mathcal{S}_2 \cup \mathcal{S}^c, i' \in \mathcal{S}_2$$

In the imputation stage, we rely only on the information from $\mathcal{S}_2$:

$$\begin{aligned} \hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{|\mathcal{S}_2|+2}}{\text{argmin}} \sum_{\substack{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}, \\ (i', j') \neq (i, j)}} \left(A_{i'j'} - \hat{\Pi}_{i'j'} - \alpha_{i'} - \alpha_{j'} \right)^2 K_h(\hat{d}_{ii'}) K_h(\hat{d}_{jj'}) \\ \hat{A}_{ij} = \hat{\alpha}_i + \hat{\alpha}_j + \hat{\Pi}_{ij} \end{aligned}$$

An practical issue concerns the choice of the tuning parameter h . We select h using a leave-one-out

cross-validation procedure. The details cross-fitting and cross validation are also discussed later. The asymptotic analysis presented later is based on the sample-splitting version, and a detailed summary of the algorithm incorporating cross-fitting and cross-validation is provided in a later section.

3 Theory for Imputation

Our method accommodates both a low-rank graphon, where the support of ζ is finite, and a continuous graphon, where the support of ζ is a continuum and the graphon function f is smooth. In what follows, we focus on the continuous case, while noting that the same arguments apply, with minor modifications, to the low-rank setting. A detailed discussion of the low-rank case is provided in the Appendix. We begin with introducing the regularity conditions.

Assumption 1 (Regularity Conditions) *Suppose that*

- (i) (**Asymptotics**) *Large N, n asymptotics with $N, n \rightarrow \infty$.*
- (ii) (**Independence**) *$\{(X_i, \xi_i)\}_{i=1}^N$ are i.i.d., and conditional on (X, ξ) , $\{A_{ij}\}_{1 \leq i < j \leq N}$ are generated independently according to (2.1).*
- (iii) (**Random sampling**) *Let $D_i \in \{0, 1\}$ indicates whether i is sampled. Assume $D_i \perp (A, X, \xi)$ for all i , and $\{D_i\}_{i=1}^N$ are independent across i .*
- (iv) (**Compact support**) *Let $\zeta_i = (X_i, \xi_i)$ and $d_\xi = d_X + d_\xi$, assume that the support of ζ_i is compact on $\mathbb{R}^{d_\xi}$. In addition, there exist constants $0 < \underline{c} < \bar{c} \leq \infty$ such that for any $\zeta \in \text{supp}(\zeta)$ and any $h > 0$, $\underline{c}h^{d_\xi} \leq \mathbb{P}(\|\zeta_i - \zeta\| \leq h) \leq \bar{c}h^{d_\xi}$.*
- (v) (**Smoothness**) *For every fixed $\zeta_j \in I$, the mapping $\zeta \mapsto f(\zeta, \zeta_j)$ is twice continuously differentiable, its second derivative is uniformly bounded for all $\zeta, \zeta_j \in \text{supp}(\zeta)$.*

Assumption 1(i) indicates that our method is designed for large-network settings where both the network size N and the number of sampled individuals n grow large. Our analysis does not rely on n and N growing at the same rate and continue to hold when n grows more slowly, e.g. $n = O(N^\alpha)$ for $\alpha \in (0, 1)$.

Assumption 1(ii) imposes that $\{(X_i, \xi_i)\}_{i=1}^N$ are i.i.d., and the data generating process of A follows a conditionally independent dyadic model (2.1). The model also follows from the Aldous-Hoover Theorem (Aldous (1981); Hoover (1979)), which shows that any infinitely exchangeable random graph admits such a representation. As discussed earlier, although (2.1) is flexible enough to encompass a wide range of network formation models commonly used in economics and statistics, it rules out strategic network formation, in which individuals linking decisions depend on those of others; see de Paula (2020) for a review.

Assumption 1(iii) is a key feature of standard egocentric sampling design, which assumes that individuals are randomly drawn from the population independently of (A, X, ξ) . This assumption can be relaxed, since our method remains valid when the observed subnetwork has a block structure, as shown in (2.3). For example, D_i may depend arbitrarily on (X_i, ξ_i) , as long as $\epsilon_{ij} \perp (D_i, D_j)$ conditional on (X_i, ξ_i, X_j, ξ_j) .

The first part of Assumption 1(iv) imposes the boundedness of ζ , and the second part introduces a standard bounded-density condition. Intuitively, it ensures that the distribution of ζ_i is neither locally too sparse nor too concentrated by requiring that for any point ζ and a small radius $h > 0$, the probability that ζ_i falls within a ball of radius h centered at ζ is of the same order as the volume of that ball. This requirement is automatically satisfied when the support is convex and the probability density function of

ζ is uniformly bounded above and bounded away from zero. This assumption is crucial for establishing the asymptotic properties of nonparametric kernel estimators.

Assumption 1(v) requires the graphon f to be twice continuously differentiable with uniformly bounded second derivatives. This smoothness condition is required because our two-way fixed-effect imputation method relies on the second-order Taylor expansion of $\Pi_{ij}^\perp$ within a local neighborhood to eliminate the first-order approximation bias, thereby improving the accuracy. Methods that rely on higher-order Taylor expansions, such as the local PCA approach proposed by Feng (2023), correspondingly require the existence of higher-order derivatives. We discuss this point in detail later.

It should be noted that the requirement of second-order differentiability of f is not a mild condition. For instance, consider the network formation model with degree heterogeneity in Example 2. If the distance function $\omega(\cdot, \cdot)$ is defined by the absolute distance, i.e., $\omega(X_i, X_j) = |X_i - X_j|$, which is quite common in applied research, the corresponding graphon f is only Lipschitz continuous rather than twice differentiable. Nevertheless, our method remains valid and can consistently impute the conditional probability of link formation, although the approximation error no longer attains the second-order rate.

3.1 Asymptotic analysis of pseudo-max distance

We now develop the asymptotic properties of the pseudo-max distance $\hat{d}_{ii'}$ and show that a sufficiently small $\hat{d}_{ii'}$ implies that ζ_i and $\zeta_{i'}$ are close in the latent space.

In the following lemma, we establish the convergence of $\hat{d}_{ii'}$ to its population counterpart $d_{ii'}$, defined as

$$d_{ii'} = \sup_{\zeta_k \in \text{supp}(\zeta)} \left| \int_{\zeta_\ell} g(\zeta_k, \zeta_\ell) (g(\zeta_i, \zeta_\ell) - g(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right|$$

Lemma 1 *Under Assumption 1, there exists constant $\gamma_1, \gamma_2 > 0$ such that*

$$\mathbb{P} \left(\max_{i \in \mathcal{S}_2 \cup \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - d_{ii'}| \geq \gamma_1 \frac{\log N}{N^{1/d_\zeta}} + \gamma_2 \frac{\log N}{\sqrt{n}} \right) \leq 1 - n^{-\frac{1}{2}}$$

Lemma 1 shows that the approximation error of $\hat{d}_{ii'}$ relative to its population counterpart $d_{ii'}$ can be decomposed into two components. The first term, $\gamma_1 \frac{\log N}{N^{1/d_\zeta}}$, captures the matching discrepancy in the latent characteristics ζ_i . Its rate deteriorates with the dimension of the latent space d_ζ , which is consistent with the well-known ‘‘curse of dimensionality’’ results in the matching literature. The second term, $\gamma_2 \frac{\log N}{\sqrt{n}}$, reflects the effect of the idiosyncratic error ϵ_{ij} . It shows that although the true graphon is unobservable, the pseudo-max distance can asymptotically denoise the observed network. The matching discrepancy error dominates the effect of idiosyncratic errors when $N \sim n$ and $d_\zeta \geq 3$, which is the common case in practice. Finally, for notational simplicity, let $\delta_{N,n} := \gamma_1 \frac{\log N}{N^{1/d_\zeta}} + \gamma_2 \frac{\log N}{\sqrt{n}}$ and henceforth focus on the event under which $|\hat{d}_{ii'} - d_{ii'}|$ is uniformly bounded by $\delta_{N,n}$, as implied by Lemma 1.

The following assumption is critical and requires that the population max-pseudo distance $d_{ii'}$ be sufficiently informative to reveal the latent distance $\|\zeta_i - \zeta_{i'}\|$.

Assumption 2 (Informativeness) *For all $\zeta_i, \zeta_{i'} \in \text{supp}(\zeta)$, there exists constant $\lambda > 0$ such that $d_{ii'} \geq \lambda \|\zeta_i - \zeta_{i'}\|$.*

This assumption ensures that when the population max-pseudo distance $d_{ii'}$ is sufficiently small, the latent positions ζ_i and $\zeta_{i'}$ are close to each other. Hence, $d_{ii'}$ can be used to identify individuals with similar latent characteristics.

Example 1 (Continued) When each individual belongs to one of G groups, $g \in \{1, \dots, G\}$, Assumption 2 can be guaranteed as long as there is heterogeneity across different groups, that is, no two groups share identical linking patterns. A formal illustration and the proof are provided in Lemma D.1 in the Appendix. When the graphon f takes a more general low-rank form, the informativeness condition still holds under similarly mild assumptions; we refer the reader to the discussions in Feng (2023) and Deaner, Hsiang and Zelenev (2025) for further details.

Example 2 (Continued) In Example 2, the latent type ζ may lie on a continuous support, and the graphon f is not necessarily of low rank. Therefore, verifying Assumption 2 in this setting is generally challenging. Nevertheless, we show that the informativeness assumption holds in the following model that serves as a benchmark in empirical applications. Specifically, Assumption 2 holds when (i) U_{ij} in (2.2) follows a Type-I extreme value distribution, and (ii) the model allows for degree heterogeneity and homophily effects on both observable and unobservable characteristics, where homophily can be captured by either absolute or quadratic distance. This finding implies that the informativeness assumption is empirically plausible and unlikely to impose restrictive constraints in applied network analysis. A formal illustration and the proof are provided in Lemma D.2 in the Appendix.

Remark An alternative to the pseudo-max distance is the Euclidean distance. More precisely, for any pair of nodes i and i' , define the squared Euclidean distance as $\hat{r}_{ii'}^2 = \frac{1}{|S_1|} \sum_{\ell \in S_1} (A_{i\ell} - A_{i'\ell})^2$. The main drawback of using the squared Euclidean distance is that it does not automatically denoise the observed network, because $\hat{d}_{ii'}^2$ includes the inner products of each error vector with itself, $\frac{1}{|S_1|} \sum_{\ell \in S_1} \epsilon_{i\ell}^2 + \frac{1}{|S_1|} \sum_{\ell \in S_1} \epsilon_{i'\ell}^2$, which do not vanish even as $n \rightarrow \infty$. In contrast, the pseudo-max distance involves cross-inner products of independent error vectors, so that the influence of noise decreases at the $\sqrt{n}$ rate, thereby asymptotically denoising the adjacency matrix.

Another alternative to the pseudo-max distance is the difference in average degrees, defined as $\hat{\rho}_{ii'} = \frac{1}{|S_1|} \sum_{\ell \in S_1} A_{i\ell} - \frac{1}{|S_1|} \sum_{\ell \in S_1} A_{i'\ell}$. Although the average-degree distance can also denoise the adjacency matrix, it requires a stronger informativeness assumption than Assumption 2. In particular, using average-degree distance requires the population average degree,

$$\zeta \mapsto \int_{\zeta'} f(\zeta, \zeta') \mathbb{P}(\zeta')$$

be a strictly monotonic function of ζ . This monotonicity requirement corresponds to the injectivity condition in Bonhomme, Lamadon and Manresa (2022). However, the condition does not hold in either Example 1 or Example 2 and is stronger than the assumption adopted in this paper.

3.2 Asymptotic analysis of imputation

Since the imputation procedure involves two parts, a nonparametric dyadic regression and a two-way fixed-effect imputation, the imputation error can accordingly be decomposed into two parts. Our analysis focuses on the second part arising from the two-way fixed-effect imputation, as the first-stage nonparametric estimator is generally much more accurate. The asymptotic properties of the first-stage estimator have been well established in the literature (see Graham, Niu and Powell (2021)).

Assumption 3 The estimation error of $\hat{\Pi}_{ij}$ admits the following uniform bound: there exist constants $\gamma_3 > 0$ such that

$$\sup_{1 \leq i, j \leq N} \mathbb{E} \left(\hat{\Pi}_{ij} - \Pi_{ij} \right) \leq n^{\frac{-2}{4+d_X}} \log(N), \quad \sup_{1 \leq i, j \leq N} \mathbb{V}^{1/2} \left(\hat{\Pi}_{ij} \right) \leq n^{\frac{-2}{4+d_X}} \log^2(N)$$

Assumption 3 states that the first-stage nonparametric dyadic regression estimator admits uniform bounds on its bias and standard deviation. Up to logarithmic factors, both bounds attain the rate $n^{-2/4+d_X}$, which is the optimal mean-squared-error rate for a twice continuously differentiable target function with d_X -dimensional covariates. Under Assumption 1, [Graham, Niu and Powell \(2021\)](#) shows that the Nadaraya-Watson estimator with a higher-order kernel satisfies Assumption 3.

Remark First, although the dyadic conditional expectation $\Pi(\cdot, \cdot)$ takes $2d_X$ arguments, the nonparametric convergence rate in Assumption 3 behaves as if Π depended on only d_X arguments, consistent with [Graham, Niu and Powell \(2021\)](#). Second, as noted by [Graham, Niu and Powell \(2021\)](#), when the entire network is observed, while the regression uses $O(N^2)$ dyads, the effective sample size is only $O(N)$ because the estimator is a U-statistic. However, the effective sample size is of order n (not $\sqrt{Nn}$) as stated in Assumption 3, and the rate is therefore solely governed by n . This argument can be verified via Hoeffding decomposition of the estimator.

Assumption 4 Suppose that the kernel $K : \mathbb{R} \mapsto \mathbb{R}_+$ is bounded and supported on $[-1, 1]$. In addition, $K(0) > 0$ and $K(\cdot)$ is Lipschitz continuous with constant $\bar{K} > 0$.

This assumption imposes regularity conditions on kernel K in the second step. The requirement standard in the literature, and many commonly used kernels, including the Epanechnikov kernel, satisfy Assumption 4.

We now briefly explain why $\hat{d}_{ii'}$ can be viewed as a noisy measure of $\|\zeta_i - \zeta_{i'}\|$ and can thus serve as its proxy in the kernel function. First, let h denote the bandwidth of the kernel function. When imputing $\Pi_{ij}^\perp$, Assumption 4 allows us to restrict attention to links (i', j') for which both $\hat{d}_{ii'} \leq h$ and $\hat{d}_{jj'} \leq h$. Second, by Lemma 1, we have $|\hat{d}_{ii'} - d_{ii'}| \leq \delta_n$ with high probability. When $\delta_n/h \rightarrow 0$, the approximation error in $\hat{d}_{ii'}$ is asymptotically negligible relative to the bandwidth h , so $d_{ii'}$ can be replaced by $\hat{d}_{ii'}$ without loss of first-order accuracy. Furthermore, under Assumption 2, $d_{ii'}$ is informative about the latent distance in the sense that $\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1}d_{ii'}$. Combining these results yields, with probability approaching one, $\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1}h$, which implies that using pseudo-max distance in the kernel function selects pairs whose latent distances lie within a $\lambda^{-1}h$ -neighborhood when imputing missing links.

The following theorem constitutes the main theoretical result of this paper.

Theorem 2 (Imputation errors) Under Assumption 1, 2, 3, 4, when $\delta_{N,n}/h \rightarrow 0$, $nh^{d_\zeta} \rightarrow \infty$, we have

$$\max_{i,j \in \mathcal{S}^c} \mathbb{E}(\hat{A}_{ij} - f_{ij}) = O(h^2), \quad \max_{i,j \in \mathcal{S}^c} \mathbb{V}^{1/2}(\hat{A}_{ij}) = O\left(\frac{\log N}{\sqrt{nh^{d_\zeta}}} + \delta_{N,n}h\right)$$

Theorem 2 establishes entry-wise uniform error bounds, rather than a Frobenius-norm bound for a submatrix. In contrast to the graphon estimation literature, i.e., [Gao, Lu and Zhou \(2015\)](#) and [Xu \(2018\)](#), our result provides guarantees for link-level prediction of the underlying graphon values.

Following the classical nonparametric regression literature, we decompose the estimation error into bias and variance components. The bias term is of order h^2 , because our local two-way fixed-effects estimator removes the first-order bias, leaving only the second-order component. The variance consists of two parts. The first term, $\log N/(nh)$, matches the rate in the standard nonparametric literature. The second term, $\delta_{N,n}h$, is unique to our estimator and arises from the use of the max-pseudo distance as a substitute for the unobserved latent factors ζ_i and ζ_j . When the latent factors are observable, then $\delta_{N,n} = 0$, and the second term disappears, reducing our result to cases [Tsybakov \(2008\)](#) and [Graham, Niu and Powell \(2021\)](#).

The bandwidth h governs the bias-variance trade-off. As h decreases, the bias diminishes; however, because our variance includes two components, the trade-off deviates from the classical setting. While

$\frac{\log N}{\sqrt{nh^{d_\zeta}}}$ increases as h shrinks, the term $\delta_{N,n}h$ decreases. Therefore, the optimal choice of h depends jointly on the relative magnitudes of n and N and on d_ζ . When n and N are of the same order, one can still set $h \asymp n^{-1/(4+d_\zeta)}$ to attain the minimum mean squared error.

Remark Our method employs two-way fixed-effects model for imputation, which effectively removes the first-order approximation error. The intuition behind this approximation can be illustrated by a Taylor expansion. Heuristically, to impute A_{ij} , if we ignore the first-step nonparametric dyadic regression and consider Taylor expansion of $f(\zeta_{i'}, \zeta_{j'})$ around fixed (ζ_i, ζ_j) :

$$\begin{aligned} f(\zeta_{i'}, \zeta_{j'}) &= \underbrace{f(\zeta_i, \zeta_j) + (\nabla_{\zeta_i} f(\zeta_i, \zeta_j))' (\zeta_{i'} - \zeta_i)}_{\alpha_{i'}} + \underbrace{(\nabla_{\zeta_j} f(\zeta_i, \zeta_j))' (\zeta_{j'} - \zeta_j)}_{\alpha_{j'}} + O(\|\zeta_{i'} - \zeta_i\|^2 + \|\zeta_{j'} - \zeta_j\|^2) \\ &= \alpha_{i'} + \alpha_{j'} + O(\|\zeta_{i'} - \zeta_i\|^2 + \|\zeta_{j'} - \zeta_j\|^2) \end{aligned}$$

Because we employ a kernel-weighted estimator based on the pseudo distances $\hat{d}_{ii'}, \hat{d}_{jj'}$, with the kernel bandwidth h , and since, as discussed above, $\hat{d}_{ii'}$ and $\hat{d}_{jj'}$ differ from the latent-space distances $\|\zeta_{i'} - \zeta_i\|$ and $\|\zeta_{j'} - \zeta_j\|$ only up to a bounded factor, the contribution to imputation comes from links (i', j') with $\hat{d}_{ii'}, \hat{d}_{jj'} \leq h$. For those links, we obtain

$$f(\zeta_{i'}, \zeta_{j'}) = \alpha_{i'} + \alpha_{j'} + O(h^2)$$

Hence, the two-way fixed-effects structure absorbs all first-order approximation errors, leaving only the second-order remainder $O(h^2)$.

The idea of using two-way fixed-effects imputation is closely related to those developed in [Freeman and Weidner \(2023\)](#) and [Beyhum and Mugnier \(2024\)](#). Our approach differs from theirs in two aspects. First, their methods do not involve missing data and are primarily designed to estimate structural parameters in panel data models, whereas our method aims to predict the formation probability of missing links. Second, they discretize the data into groups based on k-means clustering, while we employ a kernel estimator.

The methods most conceptually related to ours are those of [Feng \(2023\)](#) and [Deaner, Hsiang and Zelenev \(2025\)](#). [Deaner, Hsiang and Zelenev \(2025\)](#) propose a local-constant estimator to impute potential outcomes. Because they assume only that the potential outcome function is Lipschitz continuous, their approximation error is of order $O(h)$. Building on the ideas of [Freeman and Weidner \(2023\)](#) and [Beyhum and Mugnier \(2024\)](#), we assume that the graphon is twice continuously differentiable and, by using a two-way fixed-effects imputation, we reduce the approximation error to $O(h^2)$. In comparison, the local PCA method proposed by [Feng \(2023\)](#) leverages factor models together with higher-order Taylor expansions to achieve more accurate approximations. In particular, [Feng \(2023\)](#) shows that the local PCA estimator attains Stone's minimax-optimal rate for nonparametric regression. We refer interested readers to that paper for further details.

We adopt the two-way fixed-effects model instead of local PCA approach for three reasons. First, local PCA method requires prior knowledge of the dimension d_ζ , which is typically unknown in practice and rarely supported by prior information. In contrast, our approach does not require prior knowledge of d_ζ . Second, the higher-order smoothness assumption is often violated in empirically relevant settings. As discussed earlier, in many applied network formation models, the homophily effect depends on the absolute distance between characteristics, implying that the graphon is only Lipschitz continuous. Our approach is robust to such a lack of smoothness and can still achieve first-order approximation accuracy in this case. Third, networks in most economic applications are relatively small. For instance, in development economics, village networks typically contain around 200 households, and with a sampling

rate of roughly 40%, the effective sample size available for imputing a specific missing link can be fewer than 20. Under such small-sample conditions, achieving theoretical higher-order accuracy is difficult in practice. Our numerical simulations also show that the performance of our method is comparable to that of the local PCA approach.

4 Theory for Downstream Estimation

We consider the application of our method to estimating parameters in downstream economic models. The data $\{(A_m, X_m, W_m, Y_m)\}_{m=1}^M$ consist of M large i.i.d. networks indexed by $m = 1, 2, \dots, M$, and each network has size N_m . Here, $A_m \in \{0, 1\}^{N_m \times N_m}$ is the adjacency matrix, $Y_m \in \mathbb{R}^{N_m}$ is the outcome variable, $X_m \in \mathbb{R}^{N_m \times d_X}$ denotes the covariates that directly enter the network formation model f as in (2.1), whereas $W_m \in \mathbb{R}^{N_m \times d_W}$ contains variables that do not necessarily appear in the network formation model but affect the outcome Y_m . We note that W_m and X_m may overlap, i.e., $W_m \cap X_m \neq \emptyset$.

The adjacency matrix A_m is not fully observed because of egocentric sampling, but the outcome Y_m and the covariates (X_m, W_m) are observed for all individuals in the network. Let n_m be the number of sampled individuals in network m . We mainly focus on asymptotics in which $M, N_m, n_m \rightarrow \infty$. The large network asymptotics $N_m, n_m \rightarrow \infty$ are required to consistent imputation of the adjacency matrix, but letting $M \rightarrow \infty$ is not necessary; our analysis can also be applied to one large network.

4.1 Consistency

GMM We work with the moment function

$$\psi(A_m, X_m, W_m, Y_m, \alpha) : [0, 1]^{N_m \times N_m} \times \mathbb{R}^{N_m \times d_X} \times \mathbb{R}^{N_m \times d_W} \times \mathbb{R}^{d_\alpha} \mapsto \mathbb{R}^q$$

where q is the number of moment conditions and $\alpha \in \mathbb{R}^{d_\alpha}$ is the structural parameter of interest in the downstream analysis. We assume the following moment condition:

$$\mathbb{E}(\psi(A_m, W_m, Y_m, \alpha_0)) = 0 \quad (4.1)$$

where α_0 is the true parameter. Since A_m is partially observed due to sampling, we impute each sampled network separately to obtain $\{\hat{A}_m \mid m = 1, \dots, M\}$, and then use the imputed adjacency matrix to solve:

$$\hat{\alpha} = \arg \min_{\alpha \in \text{supp}(\alpha)} \left(\frac{1}{M} \sum_{m=1}^M \psi(\hat{A}_m, W_m, Y_m, \alpha) \right)' \Sigma_M \left(\frac{1}{M} \sum_{m=1}^M \psi(\hat{A}_m, W_m, Y_m, \alpha) \right) \quad (4.2)$$

where $\Sigma_M \in \mathbb{R}^{q \times q}$ is the GMM weight matrix.

Example 3 (Regression on centrality) A large literature studies the relationship between individuals' network positions (centrality) and their outcomes, for example, [Hochberg, Ljungqvist and Lu \(2007\)](#), [Cruz, Labonne and Querubin \(2017\)](#), and [Banerjee et al. \(2013\)](#). The regressions used in such studies can be written as:

$$Y_{mi} = \alpha_C + \alpha_1 \phi_{mi}(A_m) + e_{mi}, \quad i = 1, 2, \dots, N_m, \quad m = 1, 2, \dots, M \quad (4.3)$$

where $\alpha = (\alpha_C, \alpha_1)'$ is the parameter of interest, $\phi_{mi} : [0, 1]^{N_m \times N_m} \rightarrow \mathbb{R}$ computes the centrality for individuals i . For example:

- (i) **(Degree centrality)** For any matrix $\tilde{A} \in [0, 1]^{N_m \times N_m}$, if $\phi_{mi}(\tilde{A}) = \frac{1}{N_m} \sum_{j=1}^{N_m} \tilde{A}_{ij}$, then $\phi_{mi}(\cdot)$ represents the normalized degree of individual i when $\tilde{A}$ is an adjacency matrix.

(ii) (**Eigenvector centrality**) For any symmetric matrix $\tilde{A} \in [0, 1]^{N_m \times N_m}$, let $\tilde{A} = U_m D_m U_m'$ be the eigenvalue decomposition of $\tilde{A}$, where $U_m \in \mathbb{R}^{N_m \times N_m}$ is an orthonormal matrix containing the eigenvectors of $\tilde{A}$ and $D_m \in \mathbb{R}^{N_m \times N_m}$ is a diagonal matrix with eigenvalues arranged in decreasing order. Define $\phi_{mi}(\tilde{A}_m) = \sqrt{N_m} U_{m,i1}$, where $U_{m,i1}$ denotes the i -th entry of the leading eigenvector. When $\tilde{A}$ is a symmetric adjacency matrix, $\phi_{mi}(\cdot)$ corresponds to the eigenvector centrality of individual i .

$\{e_{mi} \mid m = 1, \dots, M, i = 1, \dots, N_m\}$ contains the error terms that may be correlated within networks, so inference should be clustered at the network level. We use the moment function:

$$\psi(\hat{A}_m, W_m, Y_m, \alpha) := \frac{1}{N_m} \sum_{i=1}^{N_m} (1, \phi_{mi}(\hat{A}_m))' (Y_{mi} - \alpha_C - \alpha_1 \phi_{mi}(\hat{A}_m))$$

to obtain estimates of α . When the errors are independent within each network, our results extend to the case of a single large network (i.e., $M = 1$), and we still obtain a consistent estimator.

Example 4 (Linear-in-mean peer-effect model) The linear-in-means model is widely used to study peer influence in social networks. The model is:

$$Y_m = \alpha_C + \alpha_{\bar{Y}} G_m Y_m + \alpha_W W_m + \alpha_{\bar{W}} G_m W_m + e_m \quad (4.4)$$

where $\alpha = (\alpha_C, \alpha_{\bar{Y}}, \alpha_W', \alpha_{\bar{W}}')'$, $\alpha_{\bar{Y}}$ captures the endogenous effect, and G_m is the row-normalized adjacency matrix². Since $G_m Y_m$ is endogenous due to the reflection problem (Manski (1993)), then under the identification conditions in Bramoullé, Djebbari and Fortin (2009), we construct the moment function:

$$\psi(A_m, W_m, Y_m, \alpha) := \frac{1}{N_m} Z_m' (Y_m - \alpha_C - \alpha_{\bar{Y}} G_m Y_m - \alpha_W W_m - \alpha_{\bar{W}} G_m W_m)$$

satisfies moment condition (4.1), where the instrument is $Z_m = [1, W_m, G_m W_m, G_m^2 W_m]$. When the network A_m is incomplete, we compute $\hat{G}_m$ and $\hat{Z}_m$ using imputed network $\hat{A}$ ³, and use the moment function:

$$\psi(\hat{A}_m, W_m, Y_m, \alpha) = \frac{1}{N_m} \hat{Z}_m' (Y_m - \alpha_C - \alpha_{\bar{Y}} \hat{G}_m Y_m - \alpha_W W_m - \alpha_{\bar{W}} \hat{G}_m W_m)$$

and to obtain α . Since $\{e_{mi} \mid m = 1, \dots, M, i = 1, \dots, N_m\}$ contains the error terms that may be correlated within networks, so inference should be clustered at the network level. However, when the errors are independent within each network, our results extend to the case of a single large network (i.e., $M = 1$), and we still obtain a consistent estimator.

Remark Our approach for estimating GMM parameters differs from that of Chandrasekhar and Lewis (2011). Specifically, Chandrasekhar and Lewis (2011) construct moments by integrating over the missing network data, typically using simulation to approximate (conditional) expectations. As a consequence,

²For any $i, j \in 1, \dots, N_m, N_m$, $G_{m,ij}$ is obtained through:

$$G_{m,ij} = \frac{A_{m,ij}}{\sum_{j'=1}^{N_m} A_{m,ij'}}$$

³For any $i, j \in 1, \dots, N_m, N_m$, $\hat{G}_{m,ij}$ is obtained through:

$$\hat{G}_{m,ij} = \frac{\hat{A}_{m,ij}}{\sum_{j'=1}^{N_m} \hat{A}_{m,ij'}}$$

and $\hat{Z}_m = [1, W_m, \hat{G}_m W_m, \hat{G}_m^2 W_m]$

their implementation is computationally demanding in GMM settings, such as the linear-in-mean peer-effects model, and it relies on parametric assumptions on the distribution of the error terms (see the discussion in [Chandrasekhar and Lewis \(2011\)](#) and [Boucher and Houndetoungan \(2020\)](#)). By contrast, our approach directly plugs the imputed network into the moment function $\psi(\cdot)$, avoiding simulation and distributional assumptions on the errors. On the other hand, a key advantage of [Chandrasekhar and Lewis \(2011\)](#)'s approach is that it can accommodate complex network statistics, such as average path length, that cannot be computed using a simple plug-in strategy. However, even in this case, our imputed adjacency matrix can still serve as the basis for the simulation-based GMM procedure using the theoretical framework developed in Section 4.2 of [Chandrasekhar and Lewis \(2011\)](#)

Assumption 5 (GMM) *Suppose that*

- (i) **(Asymptotics)** *Large M asymptotics with $M \rightarrow \infty$.*
- (ii) **(Independence)** *$\{(A_m, X_m, W_m, Y_m)\}_{m=1}^M$ are i.i.d. across networks.*
- (iii) **(Identification)** *There is a unique α_0 such that $\mathbb{E}(\psi(A_m, W_m, Y_m, \alpha_0)) = 0$.*
- (iv) **(Idiosyncrasy-robustness)** *Let F_m be the graphon matrix of adjacency matrix A_m , then as $N \rightarrow \infty$:*

$$\sup_{1 \leq m \leq M} \sup_{\alpha} \|\psi(F_m, W_m, Y_m, \alpha) - \psi(A_m, W_m, Y_m, \alpha)\| = o_P(1)$$

- (v) $\Sigma_M \xrightarrow{P} \Sigma$ and Σ is positive definite.
- (vi) **(Compactness and integrable)** *The support of α is compact, $\alpha \mapsto \psi(A_m, W_m, Y_m, \alpha)$ is continuous almost everywhere, and $\mathbb{E} \left(\sup_{\alpha \in \text{supp}(\alpha)} \|\psi(A_m, W_m, Y_m, \alpha)\|^2 \right) < \infty$.*
- (vii) *Let $\mathcal{S}_m$ be the set of sampled nodes in $\mathcal{A}$, then $\sup_{m \leq M} N^{-1} \|\hat{A}_{m, \mathcal{S}_m^c \mathcal{S}_m^c} - F_{m, \mathcal{S}_m^c \mathcal{S}_m^c}\|_F = o_P(1)$.*
- (viii) **(Lipschitz)** *For any $A_{m,1}, A_{m,2} \in [0, 1]^{N \times N}$, we have:*

$$\sup_{\alpha \in \text{supp}(\alpha)} \|\psi(F_{m,1}, W_m, Y_m, \alpha) - \psi(F_{m,2}, W_m, Y_m, \alpha)\| \leq L(W_m, Y_m) N^{-1} \|\hat{F}_{m1} - F_{m2}\|_F$$

$$\text{and } \mathbb{E} L^2(W_m, Y_m) < \infty.$$

Assumption 5(i) and (ii) are standard to ensure the consistency of M-estimator. However, if we are willing to assume that the error terms are independent within network, then these two assumptions are no longer needed, and our method can be used to study the case of a single large network; see the discussion in Examples 3 and 4.

Assumption 5(iii) is a standard identification condition and ensures that the moment condition does not hold at $\alpha \neq \alpha_0$. However, in the large-network asymptotic regime, weak identification becomes a concern. For example, in Example 4, if W is i.i.d. within each network and independent of the network (e.g., random treatment assignment), the instrument Z becomes asymptotically collinear, leading to an inconsistent estimator, see [Hayes and Levin \(2024\)](#) for further discussion. In the following analysis, we therefore assume that W is correlated with X to avoid the weak-identification problem.

Assumption 5(iv) implies that the difference between using graphon matrix F and the adjacency network A in the moment condition becomes asymptotically negligible as the network size N grows. This assumption is key for establishing consistency of our plug-in estimator: because $\hat{A}$ only converges to F on the missing subnetwork, our estimator is consistent only if replacing A with F leaves the

moment condition unchanged asymptotically, which is guaranteed by this assumption. Intuitively, Since A decomposes into underlying graphon matrix F , which is predictable from observed information, and an unpredictable error matrix, this assumption requires that the contribution of the unpredictable error matrix vanishes asymptotically. This condition is satisfied in many empirical applications using network data, such as the linear-in-means peer-effects model and linear regressions on network centralities (see the discussion in the following text). In these settings, researchers work with network statistics, which are aggregates of individual links and therefore naturally denoise the errors.

Assumption 5(v) and (vi) are standard conditions for establishing the consistency of M-estimators. Assumption 5(vii) states that, for each network m , the imputed network $\hat{A}_m$ is consistent for graphon matrix F_m on the missing subnetwork in the Frobenius norm. This condition is weaker than the conclusion in Theorem 4, as the latter establishes consistency in the sup norm for submatrix.

Assumption 5(viii) imposes a Lipschitz condition requiring that perturbations in F_m lead to changes in ψ up to a factor $L(W_m, Y_m)$ with finite second moment, uniformly over α . Since our parameter of interest is α and F_m serves as nuisance parameters, this assumption can be interpreted as a smoothness condition on the nuisance parameters within the framework of semi-parametric M-estimation.

Theorem 3 (Consistency of estimator) *Under Assumption 5, the estimator $\hat{\alpha}$ as in (4.2) is consistent for α_0 .*

4.2 Analysis of linear-in-mean peer-effect model

Consider the linear-in-mean peer-effect model (4.4) in Example 4, given weight matrix Σ_M , the GMM estimator is:

$$\hat{\alpha} = \left(\frac{1}{M} \sum_{m=1}^M W'_m Z_m \Sigma_m Z'_m W_m \right)^{-1} \left(\frac{1}{M} \sum_{m=1}^M W'_m Z_m \Sigma_m Z'_m Y_m \right) \quad (4.5)$$

Assumption 6 *Suppose that*

- (i) **(Exogeneity)** *The error vector e_m satisfies $\mathbb{E}(e_m | G_m, W_m) = 0$. Furthermore, we assume $e_m \perp \epsilon_m | (X_m, \xi_m, W_m)$, where $\epsilon_m := \{\epsilon_{u,ij} | 1 \leq i < j \leq N_m\}$ contains error terms in network m 's generating process.*
- (ii) **(Asymptotics)** *The network sizes $\{N_m | m = 1, \dots, N\}$ are of the same order, and sample sizes $\{n_m | m = 1, \dots, N\}$ are of the same order as well, i.e.,*

$$\sup_{M \rightarrow \infty} \inf_{1 \leq m_1, m_2 \leq M} (N_{m_1}/N_{m_2}) > 0, \quad \sup_{M \rightarrow \infty} \inf_{1 \leq m_1, m_2 \leq M} (n_{m_1}/n_{m_2}) > 0$$

- (iii) **(Non-degeneracy)** *Let $\lambda_{\min}(\cdot)$ denote the smallest singular value of a matrix, then:*

$$\lambda_{\min} \left(\mathbb{E} \left(\frac{1}{N_m} Z'_m W_m \right)' \mathbb{E} \left(\frac{1}{N_m} Z'_m W_m \right) \right) > 0$$

Assumption 6(i) first imposes the strict exogeneity of regressors, $\mathbb{E}(e_m | G_m, W_m) = 0$, which rules out correlated effects (Bramoullé, Djebbari and Fortin, 2020). We further assume that the errors in the linear-in-means peer-effects model, e_m , are conditionally independent of the errors in the network formation process, ϵ_m , which ensures that the imputation error is independent of Y_m .

Assumption 6(ii) requires that all networks have sizes of the same order and that the sample sizes within networks are also of the same order. In other words, no network or sample becomes disproportionately or disproportionately small.

Assumption 6(iii) imposes a non-degeneracy condition requiring that the smallest singular value of $\mathbb{E} \left(\frac{1}{N_m} Z'_m W_m \right)' \mathbb{E} \left(\frac{1}{N_m} Z'_m W_m \right)$ is bounded away from zero. This requirement is analogous to the relevance condition in instrumental-variable estimation, and is particularly important under large-network asymptotics. For example, if W_m is i.i.d. within each network and independent of the network (e.g., random treatment assignment), the instrument Z becomes asymptotically collinear, causing the smallest singular values of the above matrix to shrink toward zero and leading to an inconsistent IV estimator. In practice, one can let W_m to be correlated with X_m to satisfy this assumption.

Theorem 4 *Under conditions in Theorem 2, we have*

$$\sqrt{M} (\hat{\alpha} - \alpha_0 - b(h)) \xrightarrow{d} \mathcal{N}(0, \Omega + v_{N,n}(h))$$

where $b(h) = O(h^2)$, $v_{N,n}(h) = O\left(\frac{h^4}{nh^{d_\zeta}} + \delta_{N,n}^2 h^2 + \frac{1}{n}\right)$.

The theorem states that $\hat{\alpha} - \alpha_0$ is asymptotically normally distributed, but its limiting distribution is centered away from zero by an $O(h^2)$ bias term arising from the bias of nonparametric estimation error of $\hat{A}_{ij}$. The asymptotic variance can be decomposed into two components: the first, denoted by Ω , corresponds to the variance that would obtain if the network A were perfectly observed; the second, $v_{N,n}$, captures the additional variability introduced by the nonparametric estimation of $\hat{A}$.

5 Simulation Study

In this section, we examine the finite-sample properties of the proposed method using a series of numerical experiments. We compare its performance with several alternative imputation approaches, including low-rank imputation methods proposed by Bai and Ng (2021) and Li et al. (2023), and the local PCA method developed by Feng (2023). We also use numerical experiments to illustrate (1) the importance of incorporating observed covariates for improving prediction accuracy, and (2) the influence of sample splitting on finite-sample performance. We begin with introducing alternative method.

5.1 Alternative Methods

Covariate-only method This method is a dyadic nonparametric regression based imputation approach that relies solely on the observed covariates X . It is evident that when unobserved heterogeneity is present, the prediction error of this approach does not diminish as the sample size increases. We compare its numerical performance with that of our proposed method to highlight the importance of accounting for unobserved heterogeneity. We refer to this method as **X** in the following text.

Low-rank imputation When the observed matrix follows a strong-factor structure with an additive noise component and the missing entries exhibit a block-missing pattern⁴ (see (2.3)), Bai and Ng (2021) and Li et al. (2023) propose using low-rank estimation to impute the missing data. The key idea is to first recover the latent factors using the observed submatrices $A_{\mathcal{S}\mathcal{S}}$, $A_{\mathcal{S}\mathcal{S}^c}$, and $A_{\mathcal{S}^c\mathcal{S}}$, and then reconstruct the missing bottom-right submatrix $A_{\mathcal{S}^c\mathcal{S}^c}$ through the product of the estimated factors.

⁴Bai and Ng (2021) refer to this pattern as a tall-wide structure in the context of panel data, while Li et al. (2023) study an analogous setting in network data and refer to it as egocentric sampling.

When the graphon $f(\cdot, \cdot)$ is nonlinear, for instance, Example 2, the graphon matrix is typically high-rank and thus violates the low-rank assumption required by Bai and Ng (2021) and Li et al. (2023). Although their methods are not theoretically valid under such settings, we include them in our numerical experiments for comparison, given their simplicity and computational efficiency, to assess how our proposed method performs in practice relative to these benchmarks.

It is worth noting that both Bai and Ng (2021) and Li et al. (2023) assume that the true number of factors is known a priori. In practice, this information is rarely available; hence, in our numerical experiments, we select the number of factors via cross-validation. Additionally, neither of these methods incorporates observed covariates, whereas our approach explicitly accounts for them in the imputation process. We refer to this method as **LR** in the following text.

Local PCA The local PCA method proposed by Feng (2023) is suitable for high-rank graphon matrices. Specifically, it first identifies the neighbors of each node i based on a pseudo-distance using k-nearest-neighbor (kNN), and then performs low-rank imputation on the submatrix formed by node i 's neighbors to recover the missing entries. As discussed earlier, when the underlying graphon is smoother than twice continuously differentiable, the local PCA method can, in theory, achieve a higher-order approximation error. We compare our method with local PCA to assess whether this theoretical advantage persists in finite samples.

A practical concern is that the local PCA method requires prior knowledge of the latent dimension d_ζ , since this determines the rank used when performing PCA on each submatrix. However, d_ζ is typically unknown and rarely supported by prior knowledge. While Feng (2023) propose an ad hoc procedure in their numerical simulations to address this issue, we instead use cross-validation to jointly select both the number of neighbors k in the kNN step and the rank used in PCA.

It is worth noting that local PCA does not incorporate observed covariates⁵. In our numerical experiments, we therefore consider two implementations of their method: (1) **LPCA**, which ignores covariates X and applies local PCA directly to the adjacency matrix, and (2) **X-LPCA**, which first removes the variation explained by X and then applies local PCA to the residual matrix.

Local TWFE without covariate We also consider an alternative method that directly applies our local TWFE approach to the adjacency matrix without extracting the information explained by X . As discussed earlier, because the information about ζ recovered from observed data is noisy, it is unwise to ignore the observed covariates and absorb them into the latent factors. We include the numerical simulation results of this method to evaluate the importance of incorporating covariate information. We refer to this method as **LTWFE** and to our proposed method as **X-LTWFE** in the following text.

Sample splitting While our asymptotic analysis relies on sample splitting for theoretical tractability, this requirement is primarily technical. In practice, sample splitting reduces the effective sample size, and since network datasets in economics are typically limited in size, its finite-sample performance can be worse than that of the full-sample implementation. Accordingly, we implement all alternative methods without sample splitting. For our proposed method, we consider both versions, with and without sample splitting, to examine how this choice affects finite-sample performance. We refer to our method with sample splitting as **X-LTWFE-SP** in the following text.

Summary We summarize all imputation methods used in the numerical experiments below:

- (i) **(X)** Covariate-only nonparametric method that imputes missing outcomes using X alone.

⁵Feng (2023) briefly discuss how to include covariates in a linear form in their appendix.

- (ii) **(LR)** The low-rank imputation methods of [Bai and Ng \(2021\)](#) and [Li et al. \(2023\)](#).
- (iii) **(LPCA)** The local PCA method that ignores covariates.
- (iv) **(X-LPCA)** First removes the variation explained by X and then applies local PCA to the residual matrix.
- (v) **(LTWFE)** The local two-way fixed effects method that ignores covariates.
- (vi) **(X-LTWFE)** Our proposed method. It first removes the variation explained by X and then applies local two-way fixed effects method to the residual matrix.
- (vii) **(X-LTWFE-SP)** Our proposed method with sample-splitting.

5.2 Simulation for Imputation Accuracy

The network in the simulation is generated as

$$A_{ij} = \mathbf{1}(\omega(X_i, X_j)' \beta + g(\xi_i, \xi_j) - U_{ij} \geq 0) \quad (5.1)$$

$$\omega(X_i, X_j)' = ((X_{i1} - X_{j1})^2, (X_{i2} - X_{j2})^2), \quad g(\xi_i, \xi_j) = \xi_{i1} + \xi_{j1} - \frac{1}{8}(\xi_{i2} - \xi_{j2})^2$$

The dimensions of the observable and latent characteristics are $d_X = d_\xi = 2$, and $\{(X_i, \xi_i) \mid i = 1, \dots, N\}$ is i.i.d. across individuals. Covariates and latent factors are generated according to

$$X_{id} = \frac{1}{2}(\xi_{i1} + \xi_{i2}) + \epsilon_{X,id}, \quad \forall d = 1, 2$$

and

$$\xi_i \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right), \quad \epsilon_{X,i1}, \epsilon_{X,i2} \sim \text{Uniform}([-1, 1]), \quad \epsilon_{X,i1} \perp \epsilon_{X,i2}$$

The idiosyncratic shocks $\{U_{ij}\}_{1 \leq i < j \leq N}$ are independently drawn from a type-I extreme-value distribution. The coefficient vector β captures homophily effects on observables and takes values $\beta \in \{(-0.5, 0.5), (-2, -2)\}$. We vary the magnitude of β to control the sparsity of the network: for example, when $\beta = (-0.5, 0.5)$, the resulting network tends to be denser, with a larger number of links being formed.

The network size $N = 200$, and the simulations consider sampling rates $\varphi := n/N \in \{0.2, 0.3, 0.4, 0.5, 0.6, 0.8\}$ to compare the performance of the proposed and alternative methods under different sampling scenarios. We conduct $S = 1,000$ simulations in the numerical experiment, and for each simulation s we construct the imputed network $\hat{A}_s$ and compute the mean squared error (MSE) between $\hat{A}_s$ and the underlying graphon f_s on the missing subnetwork, namely $\frac{1}{(N-n)(N-n-1)} \sum_{i,j \in S^c} (\hat{A}_{s,ij} - f_{s,ij})^2$. The reported statistic is the root mean squared error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{S} \sum_{s=1}^S \frac{1}{(N-n)(N-n-1)} \sum_{i,j \in S^c} (\hat{A}_{s,ij} - f_{s,ij})^2} \quad (5.2)$$

Table 1: Simulation results

	X ($\times 0.01$)	LR ($\times 0.01$)	LPCA ($\times 0.01$)	LTWFE ($\times 0.01$)	X-LPCA ($\times 0.01$)	X-LTWFE ($\times 0.01$)	X-LTWFE-SP ($\times 0.01$)
$\beta = (-0.5, -0.5)$							
$\varphi = 0.2$	21.0	15.8	15.9	15.3	14.9	10.9	13.1
$\varphi = 0.3$	20.9	13.9	14.0	13.9	12.3	9.7	11.3
$\varphi = 0.4$	20.8	12.4	12.5	13.0	10.8	8.9	10.4
$\varphi = 0.5$	20.8	11.3	11.3	12.3	9.8	8.4	9.7
$\varphi = 0.6$	20.8	10.3	10.4	11.9	9.2	8.1	9.3
$\varphi = 0.8$	20.6	9.2	9.2	11.0	8.3	7.5	8.6
$\beta = (-2, -2)$							
$\varphi = 0.2$	13.8	16.3	16.4	15.3	12.7	10.2	11.3
$\varphi = 0.3$	13.7	14.8	14.9	14.0	11.5	9.4	10.3
$\varphi = 0.4$	13.7	13.3	13.5	13.1	10.5	8.9	9.7
$\varphi = 0.5$	13.6	11.7	11.8	12.5	9.8	8.6	9.3
$\varphi = 0.6$	13.6	10.5	10.6	11.9	9.3	8.3	9.0
$\varphi = 0.8$	13.5	9.2	9.2	11.0	8.4	7.8	8.5

Note: We conduct Monte Carlo simulations based on 1,000 repetitions of the network formation model (5.1). The network size is $N = 200$. We set $\beta \in \{(-0.5, 0.5), (-2, -2)\}$ and sampling rates $\varphi \in \{0.2, 0.3, 0.4, 0.5, 0.6, 0.8\}$. We report RMSE as in (5.2) for the following imputation methods: (1) covariate-only nonparametric imputation that imputes missing outcomes using X alone (**X**), (2) the low-rank imputation Bai and Ng (2021) and Li et al. (2023) (**LR**), (3) the local PCA method that ignores covariates (**LPCA**), (4) the local PCA method using covariate (**X-LPCA**), (5) local two-way fixed effects method that ignores covariates (**LTWFE**), (6) the local TWFE method using covariate (**X-LTWFE**), and (7) X-LTWFE method with sample-splitting (**X-LTWFE-SP**). It is important to note that LPCA, LTWFE, X-LPCA, and X-LTWFE are all implemented without sample-splitting. RMSEs are presented in units of 0.01 for ease of reading.

We evaluate the performance of our imputation method and alternative approaches in both dense networks (e.g., $\beta = (-0.5, -0.5)$, for which the average probability of forming a link is about 34%) and sparse networks (e.g., $\beta = (-2, -2)$, for which the average probability of forming a link is about 10%). The simulation results are reported in Table 1, with RMSEs presented in units of 0.01 for ease of reading.

The first column (**X**) reports the RMSE of covariate-only method, which imputes missing links using X alone. Because this method ignores unobserved heterogeneity, its imputations are systematically biased, and the error does not vanish as the network size or sampling rate increases. Consistent with the theoretical prediction, the numerical results show that, as the sampling rate φ increases, the RMSE of the covariate-only method exhibits very limited improvement (from 0.210 to 0.206 as in the dense scenario, and from 0.138 to 0.135 in the sparse scenario).

The second column (**LR**) reports the RMSE of the low-rank imputation, and the third column (**LPCA**) reports the RMSE of the local PCA. Since the underlying graphon in (5.1) is high-rank, local PCA should, in theory, outperform the low-rank approach. However, the numerical results indicate that the RMSEs of the two methods are nearly the same across all scenarios. We conjecture that this pattern is driven by two reasons: (1) Although the graphon is not exactly low-rank, it is approximately low-rank, so the low-rank approach still provides a good approximation to the graphon matrix, especially when performance is measured by RMSE of the missing part of the adjacency matrix. (2) Local PCA relies only on the subnetwork formed by neighbors, which leads to substantial sample loss and limits its performance in finite samples. In line with our theory, the local PCA estimator outperforms the covariate-only (**X**) method in most cases, and its prediction error decreases as sampling rate increases.

The fourth column (LTWFE) reports the RMSE of the local two-way fixed effects imputation that ignores covariates. In line with the theory, its prediction error decreases as n increases. Table 1 shows that at low sampling rates, the LTWFE method performs better than local PCA, whereas once the sampling rate becomes large (e.g., when φ exceeds 40%), local PCA outperforms LTWFE. This suggests that although local PCA has better asymptotic properties, in the sense that it achieves higher-order approximation error, it requires a larger sample size to realize this advantage, whereas LTWFE is more robust in small samples. In addition, the LTWFE method is more robust to network sparsity: as shown in Table 1, its performance is nearly unaffected by changes in the sparsity of the adjacency matrix.

The fifth column (X-LPCA) and the sixth column (X-LTWFE) report the performance of LPCA and LTWFE when incorporating covariate information. These methods first extract the part of the missing links that can be explained by X , and then apply LPCA and LTWFE to the residual matrix, respectively. Compared with LPCA and LTWFE, incorporating covariate information leads to substantial improvements in prediction accuracy, particularly for X-LTWFE. The gains are especially large when the sampling rate is low or when the network is sparse. X-LPCA exhibits a similar pattern. In addition, Table 1 shows that the proposed method outperforms all alternatives in our simulations, and its performance remains robust even under sparse networks and low sampling rates, which are common in empirical applications.

The last column (X-LTWFE-SP) reports the performance of our proposed estimator with sample-splitting. As discussed earlier, although sample-splitting facilitates the asymptotic analysis, it reduces the effective sample size, and this issue can be severe in small-sample settings. In addition, it is reasonable to expect that the correlation between the pseudo-distance and the error terms diminishes. Table 1 shows that performance of proposed imputation under sample-splitting is always worse than the performance without sample-splitting, but the difference between the two methods becomes smaller as the sample size increases, that is, as the sampling rate increases.

In summary, the simulations show that our proposed method outperforms the alternative methods and is more robust across different levels of network sparsity and sampling rates. In addition, for finite samples, we recommend using the version without sample-splitting in order to achieve higher prediction accuracy. Further numerical results are provided in the Appendix.

5.3 Simulation for estimates of parameter

In this subsection, we evaluate the performance of parameter estimates in downstream analysis using the imputed network. We compare the performance of our method with the alternative methods introduced previously. We focus on two of the most commonly used types of economic analysis based on network data:

Regression on network statistics Suppose for each network $m = 1, \dots, M$, the outcome Y_{mi} is generated according to:

$$Y_{mi} = \alpha_0 + \alpha_1 \phi_{mi}(A_m) + e_{mi} \quad (5.3)$$

where $\phi_{mi} : [0, 1]^{N_m \times N_m} \rightarrow \mathbb{R}$ computes the degree centrality for node i . For example, for any matrix $\tilde{A} \in [0, 1]^{N_m \times N_m}$, if $\phi_{mi}(\tilde{A}) = \frac{1}{N_m} \sum_{j=1}^{N_m} \tilde{A}_{ij}$, then $\phi_{mi}(\cdot)$ represents the normalized degree of individual i when $\tilde{A}$ is an adjacency matrix. Another example is eigenvector centrality. For any matrix $\tilde{A} \in [0, 1]^{N_m \times N_m}$, consider the eigenvalue decomposition: A

Linear-in-mean peer effect models

6 Empirical Application

7 Conclusion

Sampled network data are common in empirical research because collecting full network information is costly, but using sampled networks can lead to biased estimates. We propose a flexible imputation method for sampled networks and show that downstream empirical analysis using the imputed network yields consistent parameter estimates.

Our method separately imputes the part of each missing link that is explained by covariates and the remaining variation not captured by them: the former via projection onto observed covariates, and the latter using local two-way fixed effects. The imputation method avoids parametric assumptions, does not rely on low-rank restrictions, and flexibly incorporates observed covariates and unobserved heterogeneity. We establish entrywise convergence rates for the imputed matrix and prove consistency of GMM estimators based on the imputed network. We further derive the convergence rate of the corresponding estimator in the linear-in-mean peer effects model. Simulations show strong performance of our method both in terms of imputation accuracy and in downstream empirical analysis. We illustrate our method with an application to the microfinance dataset of [Banerjee et al. \(2013\)](#).

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A Implementation

A.1 1

Algorithm 2 Imputation with Cross Validation

Input: Kernel $K(\cdot)$, a list of bandwidths $\{h_b\}_{b=1,\dots,B}$, incomplete network A^{obs} , covariate matrix X .

Output: Imputed network $\hat{A}$.

Step 1: Extract information explained by X

Run dyadic nonparametric regression to obtain $\hat{\Pi}_X(\cdot, \cdot)$. Also calculate $\hat{\Pi}_{X,i'j'}^\perp = A_{i'j'} - \hat{\Pi}_{X,i'j'}$ for each observed entry (i', j') , i.e., $i' \in \mathcal{S} \vee j' \in \mathcal{S}$.

Step 2: Calculate similarity

For each $i \in 1, \dots, N$ and $i' \in \mathcal{S}$, calculate the pseudo distance using observed adjacency matrix:

$$\hat{d}_{ii'} = \max_{k=1,\dots,n} \left| \frac{1}{n} \sum_{\ell=1}^n A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|$$

Step 3: Leave-one-out cross validation

For each $h_b \in \{h_1, \dots, h_B\}$, and for each $i \in \mathcal{S}$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha}(h_b) \in \underset{\alpha \in \mathbb{R}^{n+1}}{\text{argmin}} \sum_{i' \in \mathcal{S}, j' \in \mathcal{S} \cup \{j\} \setminus \{i\}} \left(\hat{\Pi}_{X,i'j'}^\perp - \alpha_{i'} - \alpha_{j'} \right)^2 K_{h_b}(\hat{d}_{ii'}) K_{h_b}(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij}(h_b) = \hat{\alpha}_i(h_b) + \hat{\alpha}_j(h_b) + \hat{\Pi}_{X,ij}$. Truncate $\tilde{A}_{ij}(h_b)$ to get

$$\hat{A}_{ij}(h_b) = \begin{cases} \tilde{A}_{ij}(h_b), & \text{if } \tilde{A}_{ij}(h_b) \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij}(h_b) > 1 \\ 0 & \text{if } \tilde{A}_{ij}(h_b) < 0 \end{cases}$$

Then the optimal h^* solves

$$h^* = \underset{h_b \in \{h_1, \dots, h_B\}}{\text{argmin}} \sum_{i \in \mathcal{S}, j \in \mathcal{S}^c} \left(A_{ij} - \hat{A}_{ij}(h_b) \right)^2$$

Step 4: Local imputation

For each $i \in \mathcal{S}^c$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{n+2}}{\text{argmin}} \sum_{i' \in \mathcal{S} \cup \{i\}, j' \in \mathcal{S} \cup \{j\}} \left(\hat{\Pi}_{X,i'j'}^\perp - \alpha_{i'} - \alpha_{j'} \right)^2 K_{h^*}(\hat{d}_{ii'}) K_{h^*}(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij} = \hat{\alpha}_i + \hat{\alpha}_j + \hat{\Pi}_{X,ij}$. Truncate $\tilde{A}_{ij}$ to get

$$\hat{A}_{ij} = \begin{cases} \tilde{A}_{ij}, & \text{if } \tilde{A}_{ij} \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij} > 1 \\ 0 & \text{if } \tilde{A}_{ij} < 0 \end{cases}$$

and zero the diagonal of $\hat{A}$ such that $\hat{A}_{ii} = 0$ for all i .

Algorithm 3 Imputation with Cross Validation and Sample Splitting

Input: Kernel $K(\cdot)$, a list of bandwidths $\{h_b\}_{b=1,\dots,B}$, incomplete network A^{obs} , covariate matrix X .

Random split $\mathcal{S}$ into two parts, $\mathcal{S}_1, \mathcal{S}_2$, with equal sizes.

Output: Imputed network $\hat{A}$.

Step 1: Extract information explained by X

Run dyadic nonparametric regression to obtain $\hat{\Pi}_X(\cdot, \cdot)$. Also calculate $\hat{\Pi}_{X,i'j'}^\perp = A_{i'j'} - \hat{\Pi}_{X,i'j'}$ for each observed entry (i', j') , i.e., $i' \in \mathcal{S} \vee j' \in \mathcal{S}$.

Step 2: Calculate similarity

Calculate the pseudo distance using observed adjacency matrix:

$$\hat{d}_{ii'} = \max_{k \in \mathcal{S}_1} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|, \quad i \in \mathcal{S}_2 \cup \mathcal{S}^c, i' \in \mathcal{S}_2$$

Step 3: Leave-one-out cross validation

For each $h_b \in \{h_1, \dots, h_B\}$, and for each $i \in \mathcal{S}_2$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha}(h_b) \in \underset{\alpha \in \mathbb{R}^{|\mathcal{S}_2|+1}}{\text{argmin}} \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2 \cup \{j\} \setminus \{i\}} \left(\hat{\Pi}_{X,i'j'}^\perp - \alpha_{i'} - \alpha_{j'} \right)^2 K_{h_b}(\hat{d}_{ii'}) K_{h_b}(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij}(h_b) = \hat{\alpha}_i(h_b) + \hat{\alpha}_j(h_b) + \hat{\Pi}_{X,ij}$. Truncate $\tilde{A}_{ij}(h_b)$ to get

$$\hat{A}_{ij}(h_b) = \begin{cases} \tilde{A}_{ij}(h_b), & \text{if } \tilde{A}_{ij}(h_b) \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij}(h_b) > 1 \\ 0 & \text{if } \tilde{A}_{ij}(h_b) < 0 \end{cases}$$

Then the optimal h^* solves

$$h^* = \underset{h_b \in \{h_1, \dots, h_B\}}{\text{argmin}} \sum_{i \in \mathcal{S}_2, j \in \mathcal{S}^c} \left(A_{ij} - \hat{A}_{ij}(h_b) \right)^2$$

Step 4: Local imputation

For each $i \in \mathcal{S}^c$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{|\mathcal{S}_2|+2}}{\text{argmin}} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \left(\hat{\Pi}_{X,i'j'}^\perp - \alpha_{i'} - \alpha_{j'} \right)^2 K_{h^*}(\hat{d}_{ii'}) K_{h^*}(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij} = \hat{\alpha}_i + \hat{\alpha}_j + \hat{\Pi}_{X,ij}$. Truncate $\tilde{A}_{ij}$ to get

$$\hat{A}_{ij} = \begin{cases} \tilde{A}_{ij}, & \text{if } \tilde{A}_{ij} \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij} > 1 \\ 0 & \text{if } \tilde{A}_{ij} < 0 \end{cases}$$

and zero the diagonal of $\hat{A}$ such that $\hat{A}_{ii} = 0$ for all i .

B Consistency of Imputation

$$\sup_f \|f\|_\infty \leq \rho_f$$

Proof of Lemma 1. Recall that

$$\hat{d}_{ii'} = \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|$$

and its population counterpart:

$$d_{ii'} = \sup_{\zeta_k} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right|$$

and define

$$\tilde{d}_{ii'} = \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} f(\zeta_k, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) \right|$$

then by triangular inequality:

$$\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - d_{ii'}| \leq \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - \tilde{d}_{ii'}| + \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\tilde{d}_{ii'} - d_{ii'}|$$

Bound on $\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - \tilde{d}_{ii'}|$ Note that for i, i', k , we have:

$$\begin{aligned} & \left| \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right| - \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} f(\zeta_k, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) \right| \right| \\ & \leq \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell} (f(\zeta_i, \zeta_\ell) + f(\zeta_{i'}, \zeta_\ell)) \right| + \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} f(\zeta_k, \zeta_\ell) (\epsilon_{i\ell} - \epsilon_{i'\ell}) \right| \\ & \quad + \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell} (\epsilon_{i\ell} - \epsilon_{i'\ell}) \right| + \frac{2\rho_f^2}{|\mathcal{S}_1|} \end{aligned}$$

which implies

$$\begin{aligned} \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ij} - \tilde{d}_{ij}| & \leq \underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell} (f(\zeta_i, \zeta_\ell) + f(\zeta_{i'}, \zeta_\ell)) \right|}_{B_1} \\ & \quad + \underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} f(\zeta_k, \zeta_\ell) (\epsilon_{i\ell} - \epsilon_{i'\ell}) \right|}_{B_2} \\ & \quad + \underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell} (\epsilon_{i\ell} - \epsilon_{i'\ell}) \right|}_{B_3} + \frac{2\rho_f^2}{|\mathcal{S}_1|} \end{aligned}$$

We employ Bernstein's concentration bound, the conditional independence of $\{\epsilon_{ij} \mid i, j = 1, \dots, N\}$, and $|\mathcal{S}_1| = |\mathcal{S}_2| = n/2$ to show that for any $t > 0$

$$\begin{aligned}\mathbb{P}(B_1 \geq t) &\leq N^3 \mathbb{P}\left(\left|\frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell}(f(\zeta_i, \zeta_\ell) + f(\zeta_{i'}, \zeta_\ell))\right| \geq t\right) \leq 2N^3 \exp\left(-\frac{nt^2}{16\sigma^2\rho_f^2 + 8\sigma\rho_f t}\right) \\ \mathbb{P}(B_2 \geq t) &\leq N^3 \mathbb{P}\left(\left|\frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} f(\zeta_k, \zeta_\ell)(\epsilon_{i\ell} - \epsilon_{i'\ell})\right| \geq t\right) \leq 2N^3 \exp\left(-\frac{nt^2}{16\sigma^2\rho_f^2 + 8\sigma\rho_f t}\right) \\ \mathbb{P}(B_3 \geq t) &\leq N^3 \mathbb{P}\left(\left|\frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell}(\epsilon_{i\ell} - \epsilon_{i'\ell})\right| \geq t\right) \leq 2N^3 \exp\left(-\frac{nt^2}{16\sigma^4 + 8\sigma^2 t}\right)\end{aligned}$$

Then we can find a sufficiently large constant $C_1 > 0$ such that when $t = C_1 \frac{\log N}{\sqrt{n}}$, we have

$$\mathbb{P}\left(\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - \tilde{d}_{ii'}| \geq C_1 \frac{\log N}{\sqrt{n}} + \frac{4\rho_f^2}{n}\right) \leq \frac{1}{3} n^{-\frac{1}{2}}$$

Bound on $\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\tilde{d}_{ii'} - d_{ii'}|$ Note that for i, i', k , we have:

$$\begin{aligned}|\tilde{d}_{ij} - d_{ij}| &\leq \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) - \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| \\ &\quad + \sup_{\zeta_k} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| \\ &\quad - \max_{k \notin \{i, i'\}} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right|\end{aligned}$$

and $\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\tilde{d}_{ii'} - d_{ii'}|$ is bounded by:

$$\begin{aligned}&\underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) - \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right|}_{Q_1} \\ &+ \underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left(\sup_{\zeta_k} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| - \max_{k \notin \{i, i'\}} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| \right)}_{Q_2}\end{aligned}$$

One can employ Bernstein's inequality (Lemma D.3) to show that for any $t > 0$:

$$\mathbb{P}(Q_1 \geq t) \leq 2N^3 \exp\left(-\frac{nt^2}{16\rho_f^4 + 8\sigma\rho_f^4 t}\right)$$

Then we can find a sufficiently large constant $C_2 > 0$ such that when $t = C_2 \frac{\log N}{\sqrt{n}}$, we have

$$\mathbb{P}\left(Q_1 \geq C_2 \frac{\log N}{\sqrt{n}} + \frac{4\rho_f^2}{n}\right) \leq \frac{1}{3} n^{-\frac{1}{2}}$$

For Q_2 , suppose ζ^* the maximizer of $d_{ii'}$, we have

$$\begin{aligned}
S_2 &\leq \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left(\left| \int_{\zeta_\ell} f(\zeta^*, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| - \min_{k \notin \{i, i'\}} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| \right) \\
&\stackrel{(i)}{\leq} \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \min_{k \notin \{i, i'\}} 2\rho_f L d_\zeta^{1/2} \|\zeta^* - \zeta_k\| \\
&\leq 2\rho_f L d_\zeta^{1/2} \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \sup_{\zeta} \min_{k \notin \{i, i'\}} \|\zeta - \zeta_k\|
\end{aligned}$$

where inequality (i) follows from the Assumption 1(v) that since f is Lipschitz function with a constant $L > 0$. By Assumption 1(iv), there exists a t -covering of the support of ζ that consists of $\mathcal{J} \leq 4^{d_\zeta} d_\zeta^{d_\zeta/2} t^{-d_\zeta}$ open balls with radius t , denoted as $\{\bar{\zeta}_j\}_{j=1}^{\mathcal{J}}$. Therefore, we immediately have:

$$\sup_{\zeta} \min_{k \notin \{i, i'\}} \|\zeta - \zeta_k\| \leq \max_{j=1, \dots, \mathcal{J}} \min_{k \notin \{i, i'\}} \|\bar{\zeta}_j - \zeta_k\| + t$$

It follows that:

$$\begin{aligned}
\mathbb{P} \left(\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \sup_{\zeta} \min_{k \notin \{i, i'\}} \|\zeta - \zeta_k\| \geq 2t \right) &\leq N^2 \mathbb{P} \left(\max_{j=1, \dots, \mathcal{J}} \min_{k \notin \{i, i'\}} \|\bar{\zeta}_j - \zeta_k\| \geq t \right) \\
&\leq N^2 \mathcal{J} \mathbb{P} \left(\min_{k \notin \{i, i'\}} \|\bar{\zeta}_j - \zeta_k\| \geq t \right) \\
&\stackrel{(i)}{=} N^2 \mathcal{J} \left(\mathbb{P} \left(\min_{k \notin \{i, i'\}} \|\bar{\zeta}_j - \zeta_k\| \geq t \right) \right)^N \\
&\stackrel{(ii)}{\leq} N^2 \mathcal{J} (1 - \underline{c} t^{d_\zeta})^N \\
&\leq 4^{d_\zeta} d_\zeta^{d_\zeta/2} N^2 t^{-d_\zeta} (1 - \underline{c} t^{d_\zeta})^N
\end{aligned}$$

where equality (i) comes from the independence of $\{\zeta_i \mid i = 1, \dots, N\}$, and inequality (ii) follows from Assumption 1(iv). Then we can find a constant $C_3 > 0$ such that when $t = C_3 \frac{\log N}{N^{1/d_\zeta}}$,

$$\mathbb{P} \left(Q_2 \geq C_3 \frac{\log N}{N^{1/d_\zeta}} \right) \leq \frac{1}{3} n^{-\frac{1}{2}}$$

Combine all these above, and let $\gamma_1 = C_3$ as $\gamma_2 = C_1 + C_2$, then

$$\mathbb{P} \left(\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left| \hat{d}_{ij} - d_{ij} \right| \geq \gamma_1 \frac{\log N}{N^{1/d_\zeta}} + \gamma_2 \frac{\log N}{\sqrt{n}} + \frac{6\rho_f^2}{N_0} \right) \leq n^{-\frac{1}{2}}$$

and we complete the proof. It is sufficiently focusing on the event

$$\mathcal{A} := \left\{ \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left| \hat{d}_{ii'} - d_{ii'} \right| \geq \gamma_1 \frac{\log N}{N^{1/d_\zeta}} + \gamma_2 \frac{\log N}{\sqrt{n}} \right\} = \left\{ \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left| \hat{d}_{ii'} - d_{ii'} \right| \geq \delta_{N,n} \right\} \quad (\text{B.1})$$

■

Lemma B.1 *Under event $\mathcal{A}$ as in (B.1), for any $h > 0$ such that $\delta_{N,n}/h \rightarrow 0$, then for all $i \in \mathcal{S}^c, i' \in \mathcal{S}_2$, when $\hat{d}_{ii'} \leq h$, there is a constant $\eta > 0$ that*

$$\|\zeta_i - \zeta_j\| \leq \eta h$$

Proof of Lemma B.1. Note that for any $i \in \mathcal{S}^c, i' \in \mathcal{S}_2$, since $\delta_{N,n}/h \rightarrow 0$, there exists constant

$C_1 > 1$ such that

$$\{\hat{d}_{ii'} \leq h\} \subseteq \{d_{ii'} \leq h + \delta_{N,n}\} \subseteq \{d_{ii'} \leq C_1 h\}$$

Also, by Assumption 2, we have

$$\{d_{ii'} \leq C_1 h\} \subseteq \{\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1} C_1 h\}$$

Let $\eta = \lambda^{-1} C_1$, and we complete the proof. ■

Proof of Theorem 2. In the proof we focus on the event $\mathcal{A}$ as in (B.1). Since $\mathbb{P}(\mathcal{A}^c) \leq n^{-1/2}$ and both the bias and variance on $\mathcal{A}^c$ are of strictly smaller asymptotic order than those in Theorem 2, their contribution is negligible.

Bias Fixed $i, j \in \mathcal{S}^c$, for any $i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}$, for notational simplicity, define

$$\begin{aligned} K_{h,i'} &:= K(d_{ii'}/h), \quad K_{h,j'} := K(d_{jj'}/h), \quad K_{h,i'j'} := K_{h,i'} K_{h,j'} \\ \hat{K}_{h,i'} &:= K(\hat{d}_{ii'}/h), \quad \hat{K}_{h,j'} := K(\hat{d}_{jj'}/h), \quad \hat{K}_{h,i'j'} := \hat{K}_{h,i'} \hat{K}_{h,j'} \end{aligned}$$

The two-way fixed-effect estimator using weights is:

$$\begin{aligned} \hat{A}_{ij} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'j'} A_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'j'}} + \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij'} A_{ij'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} A_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}} \\ &= \underbrace{\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} f_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} + \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'} f_{ij'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} f_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}}}_{Q_1} \\ &\quad + \underbrace{\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \epsilon_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} + \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'} \epsilon_{ij'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} \epsilon_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}}}_{Q_2} \end{aligned}$$

By Assumption 1(v), the Taylor expansion of $f_{i'j'}$ reads:

$$\begin{aligned} f_{i'j'} &= f(\zeta_i, \zeta_j) + \nabla'_{\zeta_i} f(\zeta_i, \zeta_j)(\zeta_{i'} - \zeta_i) + \nabla'_{\zeta_j} f(\zeta_i, \zeta_j)(\zeta_{j'} - \zeta_j) \\ &\quad + (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' \nabla^2 f(\zeta_i + \eta'_{1,i'j'}(\zeta_{i'} - \zeta_i), \zeta_j + \eta'_{2,i'j'}(\zeta_{j'} - \zeta_j))(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j) \end{aligned}$$

where $\eta_{1,i'j'}, \eta_{2,i'j'} \in [0, 1]^{d_\zeta}$, then it is straightforward to verify that ⁶

$$\begin{aligned} Q_1 &= f(\zeta_i, \zeta_j) + \nabla'_{\zeta_i} f(\zeta_i, \zeta_j)(\bar{\zeta}_i - \zeta_i) + \hat{R}_1 \\ &\quad + f(\zeta_i, \zeta_j) + \nabla'_{\zeta_j} f(\zeta_i, \zeta_j)(\bar{\zeta}_j - \zeta_j) + \hat{R}_2 \\ &\quad - f(\zeta_i, \zeta_j) - \nabla'_{\zeta_i} f(\zeta_i, \zeta_j)(\bar{\zeta}_i - \zeta_i) - \nabla'_{\zeta_j} f(\zeta_i, \zeta_j)(\bar{\zeta}_j - \zeta_j) + \hat{R}_3 \\ &= f(\zeta_i, \zeta_j) + (\hat{R}_1 + \hat{R}_2 + \hat{R}_3) \end{aligned}$$

⁶Define

$$\bar{\zeta}_i = \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \zeta_{i'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}}, \quad \bar{\zeta}_j = \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'} \zeta_{j'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'}}$$

The remaining quadratic terms $\hat{R}_1, \hat{R}_2, \hat{R}_3$ are:

$$\begin{aligned}\hat{R}_1 &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}}, & \hat{R}_2 &= \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'}(\zeta_j - \zeta_{j'})' H_{j'}(\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'}}, \\ \hat{R}_3 &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{i'j'}(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}}\end{aligned}$$

where

$$\begin{aligned}H_{i'} &= \nabla_{\zeta_i, \zeta_i}^2 f(\zeta_i + \eta'_{1,i'j}(\zeta_{i'} - \zeta_i), \zeta_j), & H_{j'} &= \nabla_{\zeta_j, \zeta_j}^2 f(\zeta_i, \zeta_j + \eta'_{2,ij'}(\zeta_{j'} - \zeta_j)) \\ H_{i'j'} &= \nabla^2 f(\zeta_i + \eta'_{1,i'j}(\zeta_{i'} - \zeta_i), \zeta_j + \eta'_{2,ij'}(\zeta_{j'} - \zeta_j))\end{aligned}$$

Under boundedness condition of f 's second-order derivative (Assumption 1(v)), Lemma B.1, and the compact support assumption (Assumption 4), there exists $0 \leq \rho_{f''} < \infty$, such that

$$\hat{R}_1 = \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \mathbf{1}(\|\zeta_i - \zeta_{i'}\| \leq \eta h)(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} \leq \rho_{f''} \eta^2 h^2$$

By the same argument we can also show that $\hat{R}_2 \leq \rho_{f''} \eta^2 h^2$ and $\hat{R}_3 \leq 4\rho_{f''} \eta^2 h^2$. Thus,

$$\mathbb{E}(Q_1 - f_{ij} \mid \zeta) \leq 6\rho_{f''} \eta^2 h^2$$

In addition, sample-splitting guarantees that $\mathbb{E}(Q_2 \mid \zeta) = 0$, and therefore,

$$\mathbb{E}(\hat{A}_{ij} - f_{ij} \mid \zeta) \leq 6\rho_{f''} \eta^2 h^2$$

Variance Define

$$\begin{aligned}R_1 &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'}}, & R_2 &= \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j'}(\zeta_j - \zeta_{j'})' H_{j'}(\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j'}}, \\ R_3 &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'}(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{i'j'}(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'}}\end{aligned}$$

and

$$\begin{aligned}& \text{Var}(\hat{A}_{ij} - \mathbb{E}(\hat{A}_{ij} \mid \zeta)) \\ & \leq 3\text{Var}(R_1 + R_2 + R_3) + 3\text{Var}(\hat{R}_1 + \hat{R}_2 + \hat{R}_3 - R_1 - R_2 - R_3) + 3\text{Var}(Q_2)\end{aligned} \tag{B.2}$$

Step 1: Bound on $\text{Var}(R_1 + R_2 + R_3)$ For notational simplicity, define

$$g_{i'} = K_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})$$

The following inequality holds with probability $1 - 3n^{-1/2}$ by Lemma B.2:

$$\begin{aligned}
\text{Var}(R_1) &\stackrel{(i)}{\leq} \frac{\text{Var}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} (g_{i'} - \mathbb{E}(g_{i'} \mid \zeta_i))\right)}{\underline{C}_1^2 n^2 h^{2d_\zeta}} \\
&\leq \frac{\left(\mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} (g_{i'} - \mathbb{E}(g_{i'} \mid \zeta_i))\right)\right)^2}{\underline{C}_1^2 n^2 h^{2d_\zeta}} \\
&\stackrel{(ii)}{\leq} \frac{\bar{C}_3^2 n h^{d_\zeta+4} \log n}{\underline{C}_1^2 n^2 h^{2d_\zeta}} \\
&= \frac{\bar{C}_3^2 h^4 \log n}{\underline{C}_1^2 n h^{d_\zeta}}
\end{aligned} \tag{B.3}$$

where (i) and (ii) come from (B.7) and (B.9) respectively. By the same argument, we have $\text{Var}(R_2) \leq \frac{\bar{C}_3^2 h^2 \log n}{\underline{C}_1^2 n h^{d_\zeta}}$ with probability at least $1 - 3n^{-1/2}$. For notational simplicity, define:

$$g_{i'j'} = K_{h,i'j'}(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{i'j'}(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)$$

For the bound on $\text{Var}(R_3)$, note that the following inequality holds with probability $1 - 3n^{-1/2}$ by Lemma B.3:

$$\begin{aligned}
\text{Var}(R_3) &\stackrel{(i)}{\leq} \frac{\text{Var}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} (g_{i'j'} - \mathbb{E}(g_{i'j'} \mid \zeta_i, \zeta_j))\right)}{\underline{D}_1^2 n^4 h^{4d_\zeta}} \\
&\leq \frac{\left(\mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} (g_{i'j'} - \mathbb{E}(g_{i'j'} \mid \zeta_i, \zeta_j))\right)\right)^2}{\underline{D}_1^2 n^4 h^{4d_\zeta}} \\
&\stackrel{(ii)}{\leq} \frac{\bar{D}_3^2 n^3 h^{3d_\zeta+4} \log n}{\underline{D}_1^2 n^4 h^{4d_\zeta}} \\
&= \frac{\bar{D}_3^2 h^4 \log n}{\underline{D}_1^2 n h^{d_\zeta}}
\end{aligned} \tag{B.4}$$

where (i) and (ii) come from (B.14) and (B.16) respectively.

Step 2: Bound on $\text{Var}(\hat{R}_1 + \hat{R}_2 + \hat{R}_3 - R_1 - R_2 - R_3)$ Since

$$\begin{aligned}
\text{Var}(\hat{R}_1 - R_1) &= \text{Var}\left(\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'}}\right) \\
&\stackrel{(i)}{\leq} \frac{1}{\min\{\underline{C}_1, \underline{C}_2\} n^2 h^{2d_\zeta}} \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h \vee \hat{d}_{ii'} \leq h) (\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'}) (\hat{K}_{h,i'} - K_{h,i'})\right)^2 \\
&\stackrel{(ii)}{\leq} \frac{\bar{K}^2 (\delta_{N,n}/h)^2}{\min\{\underline{C}_1, \underline{C}_2\} n^2 h^{2d_\zeta}} \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h \vee \hat{d}_{ii'} \leq h) (\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})\right)^2 \\
&\stackrel{(iii)}{\leq} \frac{\bar{K}^2 (\delta_{N,n}/h)^2 \rho_{f''}^2 h^4 \max\{\eta^4, \lambda^{-4}\}}{\min\{\underline{C}_1, \underline{C}_2\} n^2 h^{2d_\zeta}} \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h \vee \hat{d}_{ii'} \leq h)\right)^2 \\
&\leq \frac{2\bar{K}^2 (\delta_{N,n}/h)^2 \rho_{f''}^2 h^4 \max\{\eta^4, \lambda^{-4}\}}{\min\{\underline{C}_1, \underline{C}_2\} n^2 h^{2d_\zeta}} \left(\mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h)\right)^2 + \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h)\right)^2\right)
\end{aligned}$$

where inequality (i) is from (B.7) and (B.8) in Lemma B.2, inequality (ii) holds because of the definition of $\mathcal{A}$, inequality (iii) follows from compact support of kernel function (Assumption 4) and Lemma B.1.

In addition, (B.7) and (B.8) in Lemma B.2 ensures that (by replacing K with uniform kernel on $[-1, 1]$) there exists constant $M > 0$ such that

$$\max \left\{ \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h) \right)^2 + \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h) \right)^2 \right\} \leq M n^2 h^{2d_\zeta}$$

Thus, we must have

$$\text{Var}(\hat{R}_1 - R_1) \leq \frac{2M\bar{K}^2(\delta_{N,n}/h)^2 \rho_{f''}^2 h^4 \max\{\eta^4, \lambda^{-4}\}}{\min\{\underline{C}_1, \underline{C}_2\}} = \frac{2M\bar{K}^2 \delta_{N,n}^2 h^2 \rho_{f''}^2 \max\{\eta^4, \lambda^{-4}\}}{\min\{\underline{C}_1, \underline{C}_2\}} \quad (\text{B.5})$$

By the same argument, we can also show that $\text{Var}(\hat{R}_2 - R_2) = O(\delta_{N,n}^2 h^2)$ and $\text{Var}(\hat{R}_3 - R_3) = O(\delta_{N,n}^2 h^2)$.

Step 3: Bound on $\text{Var}(Q_2)$ Note that for $i, j \in \mathcal{S}^c$, (1) $\{\epsilon_{i'j'} \mid i' \in \mathcal{S}_2 \cup \{i\}, i' \in \mathcal{S}_2 \cup \{i\}\}$ is independent of $\{\hat{d}_{ii'} \mid i = 1, \dots, N, i' \in \mathcal{S}_2\}$ conditional on ζ (Assumption 1(ii)), (2) $\hat{K}_{ij}$ is uniformly bounded (Assumption 4), and (3) conditional ζ , $\{\epsilon_{i'j'} \mid i' \in \mathcal{S}_2 \cup \{i\}, i' \in \mathcal{S}_2 \cup \{i\}\}$ is a set of independent sub-Gaussian random variables with $\sigma = 1$, (4) $\text{Var}(\hat{K}_{h,i'\epsilon_{i'j}} \mid \zeta) = O(h^\zeta)$, $\text{Var}(\hat{K}_{h,i'j'\epsilon_{i'j'}} \mid \zeta) = O(h^{2d_\zeta})$, then we can employ Bernstein inequality to show there exists constant $\bar{M}$ such that for any $t_1, t_2 > 0$:

$$\begin{aligned} \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'\epsilon_{i'j}} \geq t \mid \zeta \right) &\leq \exp \left(\frac{-t^2}{\bar{M}^2 n h^{d_\zeta} + \frac{1}{3} \bar{M} t} \right) \\ \mathbb{P} \left(\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'\epsilon_{ij'}} \geq t \mid \zeta \right) &\leq \exp \left(\frac{-t^2}{\bar{M}^2 n h^{d_\zeta} + \frac{1}{3} \bar{M} t} \right) \\ \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'\epsilon_{i'j'}} \geq t \mid \zeta \right) &\leq \exp \left(\frac{-t^2}{\bar{M}^2 n^2 h^{2d_\zeta} + \frac{1}{3} \bar{M} t} \right) \end{aligned}$$

Then there exists $M_1, M_2 > 0$ such that

$$\begin{aligned} \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'\epsilon_{i'j}} \geq M_1 \sqrt{n h^{d_\zeta} \log n} \mid \zeta \right) &\leq n^{-1/2} \\ \mathbb{P} \left(\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'\epsilon_{ij'}} \geq M_1 \sqrt{n h^{d_\zeta} \log n} \mid \zeta \right) &\leq n^{-1/2} \\ \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'\epsilon_{i'j'}} \geq M_2 n h^{d_\zeta} \sqrt{\log n} \mid \zeta \right) &\leq n^{-1/2} \end{aligned} \quad (\text{B.6})$$

We can show that with probability at least $1 - 3n^{-1/2}$:

$$\begin{aligned} \text{Var} \left(\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'\epsilon_{i'j}}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} \right) &\stackrel{(i)}{\leq} \frac{1}{\underline{C}_2^2 n^2 h^{2d_\zeta}} \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'\epsilon_{i'j}} \right)^2 \\ &\stackrel{(ii)}{\leq} \frac{M_1^2 n h^{d_\zeta} \log n}{\underline{C}_2^2 n^2 h^{2d_\zeta}} \\ &= \frac{M_1^2 \log n}{\underline{C}_2^2 n h^{d_\zeta}} \end{aligned}$$

where (i) holds because of (B.8) in Lemma B.2, and (ii) comes from (B.6). It directly follows that $\text{Var} \left(\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \epsilon_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} \right) \leq \frac{M_1^2 \log n}{\underline{C}_2^2 n h^{d_\zeta}}$ with probability at least $1 - 3n^{-1/2}$. In addition, with probability at least $1 - 3n^{-1/2}$:

$$\begin{aligned} \text{Var} \left(\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} \epsilon_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}} \right) &\stackrel{(i)}{\leq} \frac{1}{\underline{D}_2^2 n^4 h^{4d_\zeta}} \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} \epsilon_{i'j'} \right)^2 \\ &\stackrel{(ii)}{\leq} \frac{M_2^2 n^2 h^{2d_\zeta} \log n}{\underline{D}_2^2 n^4 h^{4d_\zeta}} \\ &= \frac{M_2^2 \log n}{\underline{D}_2^2 n^2 h^{2d_\zeta}} \end{aligned}$$

Thus, combining these results yields:

$$\text{Var} \left(\hat{A}_{ij} - \mathbb{E}(\hat{A}_{ij} \mid \zeta) \right) \leq O \left(\frac{h^4}{n h^{d_\zeta}} + \delta_{N,n}^2 h^2 + \frac{1}{n h^{d_\zeta}} + \frac{1}{n^2 h^{2d_\zeta}} \right)$$

By the asymptotics $n h^{d_\zeta} \rightarrow 0$, we have

$$\text{Var} \left(\hat{A}_{ij} - \mathbb{E}(\hat{A}_{ij} \mid \zeta) \right) \leq O \left(\delta_{N,n}^2 h^2 + \frac{1}{n h^{d_\zeta}} \right)$$

■

Lemma B.2 *Under conditions and notations in Theorem 2, for all $i, j \in \mathcal{S}^c$, we can find constants $0 < \underline{C}_1 < \bar{C}_1 < \infty$ and $0 < \underline{C}_2 < \bar{C}_2 < \infty$ such that:*

$$\mathbb{P} \left(\underline{C}_1 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} \leq \bar{C}_1 n h^{d_\zeta} \right) \geq 1 - n^{-1/2} \quad (\text{B.7})$$

$$\mathbb{P} \left(\underline{C}_2 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \leq \bar{C}_2 n h^{d_\zeta} \right) \geq 1 - 2n^{-1/2} \quad (\text{B.8})$$

In addition, there exists constant $\bar{C}_3 > 0$ such that:

$$\begin{aligned} &\mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} (K_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'}) - \mathbb{E}(K_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'}) \mid \zeta_i)) \leq \bar{C}_3 \sqrt{n h^{d_\zeta} + 4 \log n} \right) \\ &\geq 1 - n^{-1/2} \end{aligned} \quad (\text{B.9})$$

Proof of Lemma B.2. Note that for any fixed i ,

$$\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} - \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} \mid \zeta_i \right) = \sum_{i' \in \mathcal{S}_2} (K_{h,i'} - \mathbb{E}(K_{h,i'} \mid \zeta_i))$$

By the boundedness of $K_{h,i'}$ (Assumption 4) and $\text{Var}(K_{h,i'} - \mathbb{E}(K_{h,i'} \mid \zeta_i)) = O(h^{d_\zeta})$ (Assumption 1(iv), Assumption 4), Bernstein inequality of independent random variables (Wainwright (2019)) demonstrates

that there exists a constant $M > 0$ such that for any $t > 0$, we have:

$$\mathbb{P} \left(\sum_{i' \in \mathcal{S}_2} (K_{h,i'} - \mathbb{E}(K_{h,i'} \mid \zeta_i)) \geq t \mid \zeta_i \right) \leq 2 \exp \left(\frac{-t^2}{M^2 n h^{d_\zeta} + \frac{1}{3} M t} \right)$$

It follows that we can find a sufficiently large constant C_1 such that when $t = C_1 \sqrt{n h^{d_\zeta} \log n}$:

$$\mathbb{P} \left(\sum_{i' \in \mathcal{S}_2} (K_{h,i'} - \mathbb{E}(K_{h,i'} \mid \zeta_i)) \geq C_1 \sqrt{n h^{d_\zeta} \log n} \mid \zeta_i \right) \leq n^{-1/2} \quad (\text{B.10})$$

In addition,

$$\begin{aligned} \mathbb{E}(K_{h,i'} \mid \zeta_i) &= \int K \left(\frac{d(\zeta_i, \zeta_{i'})}{h} \right) d\mathbb{P}_\zeta(\zeta_{i'}) \\ &= \int_{\{\|\zeta_{i'} - \zeta_i\| \leq \lambda^{-1} h\}} K \left(\frac{d(\zeta_i, \zeta_{i'})}{h} \right) d\mathbb{P}_\zeta(\zeta_{i'}) \\ &\stackrel{(i)}{\geq} C_2 \mathbb{P}(\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1} h \mid \zeta_i) \\ &\stackrel{(ii)}{\geq} C_2 \underline{c} (\lambda^{-1} h)^{d_\zeta} \end{aligned} \quad (\text{B.11})$$

Inequality (i) follows from the Lipschitz continuity of kernel K and $K(0) > 0$ (see Assumption 4), and inequality (ii) holds because Assumption 1(iv). By the boundedness of kernel K we can also show that there exists constant $C_3 > 0$ such that

$$\mathbb{E}(K_{h,i'} \mid \zeta_i) \leq C_3 \bar{c} (\lambda^{-1} h)^{d_\zeta} \quad (\text{B.12})$$

Combining (B.10), (B.11), and (B.12), we have

$$\mathbb{P} \left(C_2 \underline{c} n (\lambda^{-1} h)^{d_\zeta} - C_1 \sqrt{n h^{d_\zeta} \log n} \leq \sum_{i' \in \mathcal{S}_2} K_{h,i'} \leq C_3 \bar{c} n (\lambda^{-1} h)^{d_\zeta} + C_1 \sqrt{n h^{d_\zeta} \log n} \mid \zeta_i \right) \leq 1 - n^{-1/2}$$

Therefore, we can find constants $0 < \underline{C}_1 < \bar{C}_1 < \infty$ such that

$$\mathbb{P} \left(\underline{C}_1 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} \leq \bar{C}_1 n h^{d_\zeta} \mid \zeta_i \right) \geq 1 - n^{-1/2}$$

Since the inequality above holds for any ζ_i , we can drop the conditioning and obtain (B.7).

Now let us turn to the derivation of (B.8). Note that when conditional on event $\mathcal{A}$, we have

$$\begin{aligned} \left| \sum_{i' \in \mathcal{S}_2 \cup \{i\}} (\hat{K}_{h,i'} - K_{h,i'}) \right| &\stackrel{(i)}{\leq} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h) \bar{K} |\hat{d}_{ii'} - d_{ii'}|/h \\ &\stackrel{(ii)}{\leq} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h) \bar{K} \delta_{N,n}/h \end{aligned}$$

where inequality (i) holds because of Assumption 4, and inequality (ii) comes from the definition of $\mathcal{A}$ in as in (B.1). Since we require $\delta_{N,n}/n \rightarrow 0$, we can employ the same argument used in proving (B.7) to

show that there exists constants $0 < C_4 < C_5 < \infty$ such that

$$\begin{aligned} & \mathbb{P} \left(C_4 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h) \leq C_5 n h^{d_\zeta} \mid \mathcal{A} \right) \leq 1 - n^{-1/2} \\ \Rightarrow & \mathbb{P} \left(C_4 n h^{d_\zeta - 1} \delta_{N,n} \leq \left| \sum_{i' \in \mathcal{S}_2 \cup \{i\}} (\hat{K}_{h,i'} - K_{h,i'}) \right| \leq C_5 n h^{d_\zeta - 1} \delta_{N,n} \mid \mathcal{A} \right) \leq 1 - n^{-1/2} \end{aligned} \quad (\text{B.13})$$

Then combining (B.7) and (B.13), we can find $0 < \underline{C}_2 < \bar{C}_2 < \infty$ such that:

$$\mathbb{P} \left(\underline{C}_2 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \leq \bar{C}_2 n h^{d_\zeta} \mid \mathcal{A} \right) \geq 1 - n^{-1/2}$$

Therefore, we can obtain (B.8) by Lemma 1.

To prove (B.9), note that for any fixed i ,

$$\begin{aligned} & \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i \right) \\ &= \sum_{i' \in \mathcal{S}_2} (K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \mathbb{E}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i)) \end{aligned}$$

By the boundedness of $K_{h,i'}$ and $H_{i'}$ (Assumption 4 and Assumption 1(v)) and

$$\text{Var}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \mathbb{E}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i)) = O(h^{d_\zeta + 4})$$

(see Assumption 1(iv), Assumption 4), Bernstein inequality of independent random variables (Wainwright (2019)) demonstrates that there exists a constant $M > 0$ such that for any $t > 0$, we have:

$$\begin{aligned} & \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbb{E}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i) \geq t \mid \zeta_i \right) \\ & \leq 2 \exp \left(\frac{-t^2}{M^2 n h^{d_\zeta + 4} + \frac{1}{3} M t} \right) \end{aligned}$$

It follows that we can find a sufficiently large constant $\bar{C}_3$ such that when $t = \bar{C}_3 \sqrt{n h^{d_\zeta + 4} \log n}$:

$$\begin{aligned} & \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbb{E}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i) \geq \bar{C}_3 \sqrt{n h^{d_\zeta + 4} \log n} \mid \zeta_i \right) \\ & \leq n^{-1/2} \end{aligned}$$

Since the inequality above is true for any ζ_i , we can drop the conditioning and obtain (B.9). ■

Lemma B.3 *Under conditions and notations in Theorem 2, for all $i, j \in \mathcal{S}^c$, we can find constants $0 < \underline{D}_1 < \bar{D}_1 < \infty$ and $0 < \underline{D}_2 < \bar{D}_2 < \infty$ such that:*

$$\mathbb{P} \left(\underline{D}_1 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'} \leq \bar{D}_1 n^2 h^{2d_\zeta} \right) \geq 1 - n^{-1/2} \quad (\text{B.14})$$

$$\mathbb{P}\left(\underline{D}_2 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} \leq \bar{D}_2 n^2 h^{2d_\zeta}\right) \geq 1 - 2n^{-1/2} \quad (\text{B.15})$$

In addition, there exists constant $\bar{D}_3 > 0$ such that:

$$\begin{aligned} & \mathbb{P}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} (K_{h,i'j'}(\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})' H_{i'j'}(\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})) \right. \\ & \quad \left. - \mathbb{E}(K_{h,i'j'}(\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})' H_{i'j'}(\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'}) \mid \zeta_i, \zeta_j) \leq \bar{D}_3 \sqrt{n^3 h^{3d_\zeta + 4} \log n}\right) \geq 1 - n^{-1/2} \end{aligned} \quad (\text{B.16})$$

Proof of Lemma B.3. Consider the decomposition:

$$\begin{aligned} & \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'} \mid \zeta_i, \zeta_j\right) \\ &= K(0) \left(\sum_{i' \in \mathcal{S}_2} K_{h,i'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2} K_{h,i'} \mid \zeta_i\right)\right) + K(0) \left(\sum_{j' \in \mathcal{S}_2} K_{h,j'} - \mathbb{E}\left(\sum_{j' \in \mathcal{S}_2} K_{h,j'} \mid \zeta_j\right)\right) \\ & \quad + \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \left(K_{h,i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} \mid \zeta_i, \zeta_j\right)\right) \end{aligned} \quad (\text{B.17})$$

The property of the first two terms has been developed in Lemma B.2, so we focus on the third term in the following proof. Note that $K_{h,i'j'}$ is uniformly bounded and $\mathbb{E}(\mathbb{E}^2(K_{h,i'j'} \mid \zeta_i, \zeta_j, \zeta_{i'}) \mid \zeta_i, \zeta_j) = O(h^{3d_\zeta})$, then by Bernstein's inequality of U-statistics (Theorem 2 in Arcones (1995)), there exists sufficiently large constant M such that for any $t > 0$, we have

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} \mid \zeta_i, \zeta_j\right) \geq t\right) \leq 4 \exp\left(\frac{-t^2}{M^2 n^3 h^{3d_\zeta} + Mnt}\right)$$

We can find a constant D_1 such that when $t = D_1 \sqrt{n^3 h^{3d_\zeta} \log n}$,

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} \mid \zeta_i, \zeta_j\right) \geq D_1 \sqrt{n^3 h^{3d_\zeta} \log n}\right) \leq n^{-1/2} \quad (\text{B.18})$$

In addition,

$$\begin{aligned} \mathbb{E}(w_{i'j'} \mid \zeta_i, \zeta_j) &= \int K\left(\frac{d(\zeta_i, \zeta_{i'})}{h}\right) K\left(\frac{d(\zeta_j, \zeta_{j'})}{h}\right) d\mathbb{P}_\zeta(\zeta_{i'}) d\mathbb{P}_\zeta(\zeta_{j'}) \\ &= \int_{\{\|\zeta_{i'} - \zeta_i\| \leq \lambda^{-1}h\}} K\left(\frac{d(\zeta_i, \zeta_{i'})}{h}\right) d\mathbb{P}_\zeta(\zeta_{i'}) \int_{\{\|\zeta_{j'} - \zeta_j\| \leq \lambda^{-1}h\}} K\left(\frac{d(\zeta_j, \zeta_{j'})}{h}\right) d\mathbb{P}_\zeta(\zeta_{j'}) \\ &\stackrel{(i)}{\geq} D_2 \mathbb{P}(\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1}h \mid \zeta_i) \mathbb{P}(\|\zeta_j - \zeta_{j'}\| \leq \lambda^{-1}h \mid \zeta_j) \\ &\stackrel{(ii)}{\geq} D_2 \underline{c}^2 (\lambda^{-1}h)^{2d_\zeta} \end{aligned} \quad (\text{B.19})$$

Inequality (i) follows from the Lipschitz continuity of kernel K and $K(0) > 0$ (see Assumption 4), and inequality (ii) holds because Assumption 1(iv). By the boundedness of kernel K we can also show that there exists constant $D_3 > 0$ such that

$$\mathbb{E}(K_{h,i'j'} \mid \zeta_i) \leq D_3 \bar{c}^2 (\lambda^{-1}h)^{2d_\zeta} \quad (\text{B.20})$$

Combining (B.18), (B.19), and (B.20), we have

$$\begin{aligned} & \mathbb{P} \left(D_2 \bar{c}^2 n^2 (\lambda^{-1} h)^{2d_\zeta} - D_1 \sqrt{n^3 h^{3d_\zeta} \log n} \leq \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h, i'} \leq D_3 \bar{c}^2 n^2 (\lambda^{-1} h)^{2d_\zeta} + C_1 \sqrt{n^3 h^{3d_\zeta} \log n} \mid \zeta_i, \zeta_j \right) \\ & \leq 1 - n^{-1/2} \end{aligned}$$

In addition, by Lemma B.2, it is straightforward to see that the order of the first two terms in (B.17) is asymptotically negligible compared to the third term. Thus, we can find constants $0 < \underline{D}_1 < \bar{D}_1 < \infty$ such that

$$\mathbb{P} \left(\underline{D}_1 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h, i' j'} \leq \bar{D}_1 n^2 h^{2d_\zeta} \mid \zeta_i, \zeta_j \right) \geq 1 - n^{-1/2}$$

Since the inequality above holds for any ζ_i, ζ_j , we can drop the conditioning and obtain (B.14).

Now let us turn to the derivation of (B.15). Note that when conditional on event $\mathcal{A}$, we have

$$\begin{aligned} \left| \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} (\hat{K}_{h, i' j'} - K_{h, i' j'}) \right| & \stackrel{(i)}{\leq} \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \mathbf{1}(\hat{d}_{ii'} \leq h) \mathbf{1}(\hat{d}_{jj'} \leq h) \bar{K} \rho_K \frac{|\hat{d}_{ii'} - d_{ii'}| + |\hat{d}_{jj'} - d_{jj'}|}{h} \\ & \quad + \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \mathbf{1}(\hat{d}_{ii'} \leq h) \mathbf{1}(\hat{d}_{jj'} \leq h) \bar{K}^2 \frac{|\hat{d}_{ii'} - d_{ii'}| |\hat{d}_{jj'} - d_{jj'}|}{h^2} \\ & \stackrel{(ii)}{\leq} \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \mathbf{1}(\hat{d}_{ii'} \leq h) \mathbf{1}(\hat{d}_{jj'} \leq h) \left(\bar{K} \rho_K \frac{\delta_{N,n}}{h} + \bar{K}^2 \frac{\delta_{N,n}^2}{h^2} \right) \end{aligned}$$

where inequality (i) holds because of Assumption 4, and inequality (ii) comes from the definition of $\mathcal{A}$ in as in (B.1). Since we require $\delta_{N,n}/n \rightarrow 0$, we can employ the same argument used in proving (B.7) to show that there exists constants $0 < D_4 < D_5 < \infty$ such that

$$\begin{aligned} & \mathbb{P} \left(D_4 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \mathbf{1}(\hat{d}_{ii'} \leq h) \mathbf{1}(\hat{d}_{jj'} \leq h) \leq D_5 n^2 h^{2d_\zeta} \mid \mathcal{A} \right) \leq 1 - n^{-1/2} \\ \Rightarrow & \mathbb{P} \left(D_4 n^2 h^{2d_\zeta - 1} \delta_{N,n} \leq \left| \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} (\hat{K}_{h, i' j'} - K_{h, i' j'}) \right| \leq D_5 n^2 h^{2d_\zeta - 1} \delta_{N,n} \mid \mathcal{A} \right) \leq 1 - n^{-1/2} \end{aligned} \tag{B.21}$$

Then combining (B.14) and (B.21), we can find $0 < \underline{D}_2 < \bar{D}_2 < \infty$ such that:

$$\mathbb{P} \left(\underline{D}_2 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h, i' j'} \leq \bar{D}_2 n^2 h^{2d_\zeta} \mid \mathcal{A} \right) \geq 1 - n^{-1/2}$$

Therefore, we can obtain (B.15) by Lemma 1.

Let $g_{i' j'}$ to denote $K_{h, i' j'} (\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})' H_{i' j'} (\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})$ for notational simplicity. To prove (B.16), note that

$$\begin{aligned} & \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} (g_{i' j'} - \mathbb{E}(g_{i' j'} \mid \zeta_i, \zeta_j)) \\ & = \sum_{i' \in \mathcal{S}_2} (g_{i' j} - \mathbb{E}(g_{i' j} \mid \zeta_i, \zeta_j)) + \sum_{j' \in \mathcal{S}_2} (g_{i j'} - \mathbb{E}(g_{i j'} \mid \zeta_i, \zeta_j)) + \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} (g_{i' j'} - \mathbb{E}(g_{i' j'} \mid \zeta_i, \zeta_j)) \end{aligned} \tag{B.22}$$

The property of the first two terms has been developed in Lemma B.2, so we focus on the third term in the following proof. Note that by the boundedness of $K_{h, i'}$ and $H_{i'}$ (Assumption 4 and Assump-

tion 1(v)), we have that $g_{i'j'}$ is uniformly bounded and $\mathbb{E}(\mathbb{E}^2(g_{h,i'j'} \mid \zeta_i, \zeta_j, \zeta_{i'}) \mid \zeta_i, \zeta_j) = O(h^{3d_\zeta+4})$. Then by Bernstein's inequality of U-statistics (Theorem 2 in Arcones (1995)), there exists sufficiently large constant M such that for any $t > 0$, we have

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} g_{i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} g_{i'j'} \mid \zeta_i, \zeta_j\right) \geq t\right) \leq 4 \exp\left(\frac{-t^2}{M^2 n^3 h^{3d_\zeta+4} + M n h^2 t}\right)$$

It follows that we can find a sufficiently large constant D_5 such that when $t = D_5 \sqrt{n^3 h^{3d_\zeta+4} \log n}$:

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} g_{i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} g_{i'j'} \mid \zeta_i, \zeta_j\right) \geq D_5 \sqrt{n^3 h^{3d_\zeta+4} \log n}\right) \leq n^{-1/2}$$

In addition, by Lemma B.2, it is straightforward to see that the order of the first two terms in (B.22) is asymptotically negligible compared to the third term. Thus, we can find constants $\bar{D}_3 > 0$ such that

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} g_{i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} g_{i'j'} \mid \zeta_i, \zeta_j\right) \geq \bar{D}_3 \sqrt{n^3 h^{3d_\zeta+4} \log n}\right) \leq n^{-1/2}$$

Since the inequality above is true for any ζ_i, ζ_j , we can drop the conditioning and obtain (B.16). ■

C Proofs for Empirical Applications

Proof of Theorem 3.

$$\begin{aligned} \sup_{\beta} \left\| \psi(\hat{A}, X_m, Y_m, Y_m, \beta) - \mu(\beta) \right\| &\leq \sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(\hat{A}, X_m, Y_m, Y_m, \beta) - \frac{1}{M} \sum_{m=1}^M \psi(F_m, X_m, Y_m, Y_m, \beta) \right\| \\ &\quad + \sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(F_m, X_m, Y_m, Y_m, \beta) - \frac{1}{M} \sum_{m=1}^M \psi(A_m, X_m, Y_m, Y_m, \beta) \right\| \\ &\quad + \sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(A_m, X_m, Y_m, Y_m, \beta) - \mathbb{E}(\psi(A_m, X_m, W_m, Y_m, \beta)) \right\| \end{aligned}$$

Note that

$$\begin{aligned} &\sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(\hat{A}, X_m, Y_m, Y_m, \beta) - \frac{1}{M} \sum_{m=1}^M \psi(F_m, X_m, Y_m, Y_m, \beta) \right\| \\ &\leq \frac{1}{M} \sum_{m=1}^M L(X_m, W_m, Y_m) N^{-1} \|\hat{F}_{m1} - F_{m2}\|_F \\ &\stackrel{(i)}{=} o_P(1) \end{aligned}$$

where (i) is from Assumption 5(vii) and Assumption 5(viii). In addition, by Assumption 5(iv), we have

$$\sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(F_m, X_m, Y_m, Y_m, \beta) - \frac{1}{M} \sum_{m=1}^M \psi(A_m, X_m, Y_m, Y_m, \beta) \right\| = o_P(1)$$

By Assumption 5??, we have

$$\sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(A_m, X_m, Y_m, Y_m, \beta) - \mathbb{E}(\psi(A_m, X_m, W_m, Y_m, \beta_0)) \right\| = o_P(1)$$

Thus, we must have $\sup_{\beta} \left\| \psi(\hat{A}, X_m, Y_m, Y_m, \beta) - \mathbb{E}(\psi(A_m, X_m, W_m, Y_m, \beta_0)) \right\| \xrightarrow{p} 0$. Combining the uniform convergence result above, Assumption 5(iii), (v), ??, and Theorem 2.1 in Newey and McFadden (1994), we can prove that $\hat{\beta} \xrightarrow{p} \beta_0$. ■

For notational simplicity, define

$$p(\zeta) = \int f(\zeta, \zeta') d\mathbb{P}(\zeta')$$

and assume that $\inf_{\zeta} p(\zeta) > 0$.

Theorem C.1 1

Proof of Theorem C.1. Consider the linear regression model:

$$Y = \beta GW + e$$

which is a simple linear-mean regression model, and let us consider the asymptotics of GW and therefore, $\hat{\beta}$. We always assume that W is fixed or conditional on W , e are independent with each other. The OLS

estimator is

$$\begin{aligned}\hat{\beta} &= \left(W' \hat{G}' \hat{G} W\right)^{-1} W' \hat{G}' Y \\ &= \beta - \beta \left(W' \hat{G}' \hat{G} W\right)^{-1} W' \hat{G}' (\hat{G} - G) W + \left(W' \hat{G}' \hat{G} W\right)^{-1} W' (\hat{G} - G)' e + \left(W' \hat{G}' \hat{G} W\right)^{-1} W' G' e\end{aligned}$$

Note that for each $i \in \mathcal{S}^c$, we have

$$\begin{aligned}(\hat{G}W - GW)_i &= \frac{\sum_{j=1, \dots, N} A_{ij} W_j + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) W_j}{\sum_{j=1, \dots, N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})} - \frac{\sum_{j=1, \dots, N} A_{ij} W_j}{\sum_{j=1, \dots, N} A_{ij}} \\ &= \underbrace{\left(\sum_{j=1, \dots, N} A_{ij} W_j \right) \left(\frac{1}{\sum_{j=1, \dots, N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})} - \frac{1}{\sum_{j=1, \dots, N} A_{ij}} \right)}_{B_1} \\ &\quad + \underbrace{\frac{\sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) W_j}{\sum_{j=1, \dots, N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})}}_{B_2}\end{aligned}$$

By Lemma C.2, we have

$$\begin{aligned}\mathbb{E} \left(\frac{1}{\sum_{j=1, \dots, N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})} - \frac{1}{\sum_{j=1, \dots, N} A_{ij}} \right) &= \frac{1}{N} O \left(h^2 + h^4 + \frac{h^4}{n h^{d_\zeta}} + h^2 \delta_{N,n}^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right) \\ &= \frac{1}{N} O \left(h^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right)\end{aligned}$$

Since $\sum_{j=1, \dots, N} A_{ij} W_j = O(N)$, we have:

$$\mathbb{E}(B_1) = O \left(h^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right)$$

In addition,

$$\begin{aligned}\text{Var} \left(\frac{1}{\sum_{j=1, \dots, N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})} - \frac{1}{\sum_{j=1, \dots, N} A_{ij}} \right) \\ \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - p(\zeta_i)) \right) = 0, \quad \text{Var} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - p(\zeta_i)) \right) = O(N)\end{aligned}$$

$$\hat{G}_{ij} - G_{ij} = O(h^2) + \frac{1}{n h^d} \sum_{j' \in \mathcal{N}_j} \epsilon_{ij'} + \frac{1}{n h^d} \sum_{i' \in \mathcal{N}_i} \epsilon_{i'j} - \frac{1}{n^2 h^{2d}} \sum_{i' \in \mathcal{N}_i, j' \in \mathcal{N}_j} \epsilon_{i'j'}$$

■

Lemma C.2 *Under conditions in Theorem 2, we have that for any $i \in \mathcal{S}^c$:*

$$\mathbb{E} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - \hat{A}_{ij}) \right) = O(Nh^2), \quad \text{Var} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - \hat{A}_{ij}) \right) = N^2 O \left(\frac{h^4}{n h^{d_\zeta}} + h^2 \delta_{N,n}^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right) \quad (\text{C.1})$$

In addition, there exist constant $\bar{E}_1 > 0$ and $c > 0$, such that:

$$\mathbb{P} \left(\left| \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) \right| \leq \bar{E}_1 N \left(h^2 + \frac{h^2}{\sqrt{nh^{d_\zeta}}} + h\delta_{N,n} + \frac{1}{\sqrt{n}} + \frac{1}{nh^{d_\zeta}} \right) \right) \geq 1 - cn^{-1/2} \quad (\text{C.2})$$

In addition, when $\{W_i \mid i = 1, 2, \dots, N\}$ are uniformly bounded and $\mathbb{E}(\epsilon_{ij} \mid \{W_i \mid i = 1, 2, \dots, N\}) = 0$ for any i, j , we have:

$$\mathbb{E} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - \hat{A}_{ij}) W_j \right) = O(Nh^2), \quad \text{Var} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - \hat{A}_{ij}) W_j \right) = N^2 O \left(\frac{h^4}{nh^{d_\zeta}} + h^2 \delta_{N,n}^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right) \quad (\text{C.3})$$

In addition, there exist constant $\bar{E}_2 > 0$ and $c > 0$, such that:

$$\mathbb{P} \left(\left| \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) W_j \right| \leq \bar{E}_2 N \left(h^2 + \frac{h^2}{\sqrt{nh^{d_\zeta}}} + h\delta_{N,n} + \frac{1}{\sqrt{n}} + \frac{1}{nh^{d_\zeta}} \right) \right) \geq 1 - cn^{-1/2} \quad (\text{C.4})$$

Proof of Lemma C.2. For notational simplicity, we define

$$\begin{aligned} K_{h,i,i'} &:= K(d_{ii'}/h), \quad K_{h,j,j'} := K(d_{jj'}/h), \quad K_{h,ij,i'j'} := K_{h,i,i'} K_{h,j,j'} \\ \hat{K}_{h,i,i'} &:= K(\hat{d}_{ii'}/h), \quad \hat{K}_{h,j,j'} := K(\hat{d}_{jj'}/h), \quad \hat{K}_{h,ij,i'j'} := \hat{K}_{h,i,i'} \hat{K}_{h,j,j'} \end{aligned}$$

Note that for any $i, j \in \mathcal{S}^c$,

$$\hat{A}_{ij} - A_{ij} = R_{ij,1} + R_{ij,2} + R_{ij,3} + (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3}) + Q_{ij} + \epsilon_{ij}$$

where

$$\begin{aligned} R_{ij,1} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}}, \quad R_{ij,2} = \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}}, \\ R_{ij,3} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}} \\ \hat{R}_{ij,1} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}}, \quad \hat{R}_{ij,2} = \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}}, \\ \hat{R}_{ij,3} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}} \end{aligned}$$

and

$$Q_{ij} = \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'} \epsilon_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}} + \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'} \epsilon_{ij'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'} \epsilon_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}}$$

then

$$\begin{aligned} \sum_{j \in 1, \dots, N} (\hat{A}_{ij} - A_{ij}) &= \sum_{j \in \mathcal{S}^c} (R_{ij,1} + R_{ij,2} + R_{ij,3}) \\ &\quad + \sum_{j \in \mathcal{S}^c} (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3}) + \sum_{j \in \mathcal{S}^c} Q_{ij} + \sum_{j \in \mathcal{S}^c} \epsilon_{ij} \end{aligned}$$

The bias of $\sum_{j \in 1, \dots, N} (\hat{A}_{ij} - A_{ij})$ is of order at most (Nh^2) , and we focus on the variance of $\sum_{j \in 1, \dots, N} (\hat{A}_{ij} - A_{ij})$ in the following text.

Step 1: $\sum_{j \in \mathcal{S}^c} (R_{ij,1} + R_{ij,2} + R_{ij,3})$ Recall that

$$\sum_{j \in \mathcal{S}^c} R_{ij,1} = \sum_{j \in \mathcal{S}^c} \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}}$$

Note that since

$$\begin{aligned} \mathbb{E} (\mathbb{E}^2 (K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i, \zeta_j) \mid \zeta_i) &= O(h^{2d_\zeta+4}) \\ \mathbb{E} (\mathbb{E}^2 (K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i, \zeta_{i'}) \mid \zeta_i) &= O(h^{d_\zeta+4}) \end{aligned}$$

by Hoeffding decomposition, there exist large constant M such that for any $t > 0$, we have:

$$\begin{aligned} &\mathbb{P} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) - \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i \right) \geq t \mid \zeta_i \right) \\ &\leq 4 \exp \left(\frac{-t^2}{M^2 n^2 N h^{2d_\zeta+4} + M n h^2 t} \right) + 4 \exp \left(\frac{-t^2}{M^2 n N^2 h^{d_\zeta+4} + M N h^2 t} \right) \end{aligned}$$

Then there exist a constant $E_1 > 0$ such that when $t = E_1 \sqrt{n N^2 h^{d_\zeta+4} \log n}$, we have:

$$\begin{aligned} &\mathbb{P} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) - \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i \right) \right. \\ &\quad \left. \geq E_1 \sqrt{n N^2 h^{d_\zeta+4} \log n} \mid \zeta_i \right) \leq 1 - n^{1/2} \end{aligned} \tag{C.5}$$

By the same method, we can also show that there exist constant E_2, E_3 such that:

$$\begin{aligned} &\mathbb{P} \left(\sum_{j \in \mathcal{S}^c} \sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'}) - \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} \sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'}) \mid \zeta_i \right) \right. \\ &\quad \left. \geq E_2 \sqrt{n N^2 h^{3d_\zeta+4} \log n} \right) \leq 1 - n^{1/2} \end{aligned} \tag{C.6}$$

and

$$\begin{aligned} &\mathbb{P} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j) \right. \\ &\quad \left. - \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j) \mid \zeta_i \right) \right. \\ &\quad \left. \geq E_3 \sqrt{n^3 N^2 h^{5d_\zeta+4} \log n} \right) \leq 1 - n^{1/2} \end{aligned} \tag{C.7}$$

Thus, by (C.5), (C.6), (C.7), (B.8), and (B.15), we have

$$\begin{aligned}\text{Var} \left(\sum_{j \in \mathcal{S}^c} R_{ij,1} \right) &= \text{Var} \left(\sum_{j \in \mathcal{S}^c} \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}} \right) \\ &= O \left(\frac{1}{n^2 h^{2d_\zeta}} \right) O \left(N^2 (n d^{\zeta+4} \log n) \right) \\ &= O \left(\frac{N^2 h^4 \log n}{n h^{d_\zeta}} \right)\end{aligned}\tag{C.8}$$

$$\begin{aligned}\text{Var} \left(\sum_{j \in \mathcal{S}^c} R_{ij,2} \right) &= \text{Var} \left(\sum_{j \in \mathcal{S}^c} \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}} \right) \\ &= O \left(\frac{1}{n^2 h^{2d_\zeta}} \right) O \left(N^2 n h^{3d_\zeta+4} \log n \right) \\ &= \frac{N^2 h^{d_\zeta+4} \log n}{n}\end{aligned}\tag{C.9}$$

$$\begin{aligned}\text{Var} \left(\sum_{j \in \mathcal{S}^c} R_{ij,3} \right) &= \text{Var} \left(\sum_{j \in \mathcal{S}^c} \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}} \right) \\ &= O \left(\frac{1}{n^4 h^{4d_\zeta}} \right) O \left(n^3 N^2 h^{5d_\zeta+4} \log n \right) \\ &= \frac{N^2 h^{d_\zeta+4} \log n}{n}\end{aligned}\tag{C.10}$$

Step 2: $\sum_{j \in \mathcal{S}^c} (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3})$ It is easy to check that by Lemma 1 and using the same argument in the proof of Theorem 2, we can show that there exist a constant $E_4 > 0$ such that with probability at least $1 - n^{-1/2}$:

$$\sum_{j \in \mathcal{S}^c} (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3}) \leq E_4 N h \delta_{N,n}\tag{C.11}$$

$$\text{Var} \left(\sum_{j \in \mathcal{S}^c} (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3}) \right) = O \left(N^2 h^2 \delta_{N,n}^2 \right)\tag{C.12}$$

Step 3: $\sum_{j \in \mathcal{S}^c} (Q_{ij})$ Note that by the conditional independence between $\{\hat{d}_{ii'} \mid i \in \mathcal{S}_2, i' \in \mathcal{S}_2 \cup \mathcal{S}^c\}$ and $\{\epsilon_{ij} \mid i < j, i, j \in \mathcal{S}_2 \cup \mathcal{S}^c\}$, by Hoeffding inequality and the same method used in Theorem 2, we immediately have that there exist constant $E_5, E_6, E_7 > 0$ such that with probability at least $1 - n^{-1/2}$:

$$\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i' \epsilon_{ij}} \leq E_5 \sqrt{N n h^{d_\zeta} \log N}\tag{C.13}$$

$$\sum_{j \in \mathcal{S}^c} \sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j' \epsilon_{ij'}} = \sum_{j' \in \mathcal{S}_2} \left(\sum_{j \in \mathcal{S}^c} \hat{K}_{h,j,j'} \right) \epsilon_{ij'} + K(0) \sum_{j \in \mathcal{S}^c} \epsilon_{ij} \leq E_6 \sqrt{N^2 n h^{2d_\zeta} \log N}\tag{C.14}$$

$$\begin{aligned}\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j' \epsilon_{ij'}} &= \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2} \left(\sum_{j \in \mathcal{S}^c} \hat{K}_{h,ij,i'j'} \right) \epsilon_{i'j'} + K(0) \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \sum_{j \in \mathcal{S}^c} \hat{K}_{h,i,i' \epsilon_{ij}} \\ &= E_7 \sqrt{n^2 N^2 h^{2d_\zeta} \log N}\end{aligned}\tag{C.15}$$

Thus, by (C.13), (C.14), (C.15), (B.8), and (B.15), we have

$$\text{Var} \left(\frac{\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'} \epsilon_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}} \right) = O \left(\frac{N n h^{d_\zeta} \log N}{n^2 h^{2d_\zeta}} \right) = O \left(\frac{N}{n h^{d_\zeta}} \right) \quad (\text{C.16})$$

$$\text{Var} \left(\frac{\sum_{j \in \mathcal{S}^c} \sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'} \epsilon_{ij'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}} \right) = O \left(\frac{N^2 n h^{2d_\zeta} \log N}{n^2 h^{2d_\zeta}} \right) = O \left(\frac{N^2}{n} \right) \quad (\text{C.17})$$

$$\text{Var} \left(\frac{\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'} \epsilon_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}} \right) = O \left(\frac{N^2 n^2 h^{2d_\zeta} \log N}{n^4 h^{4d_\zeta}} \right) = O \left(\frac{N^2 \log N}{n^2 h^{2d_\zeta}} \right) \quad (\text{C.18})$$

Step 4 $\sum_{j \in \mathcal{S}^c} \epsilon_{ij}$ By the conditional independence of $\{\epsilon_{ij} \mid 1 \leq i < j \leq N\}$ and Hoeffding, we immediately conclude that there exist constant $E_8 > 0$ such that with probability at least $1 - n^{-1/2}$:

$$\sum_{j \in \mathcal{S}^c} \epsilon_{ij} \leq E_8 \sqrt{N \log N} \quad (\text{C.19})$$

and

$$\text{Var} \left(\sum_{j \in \mathcal{S}^c} \epsilon_{ij} \right) = O(N \log N) \quad (\text{C.20})$$

Combining (C.8), (C.9), (C.10), (C.16), (C.17), (C.18), and (C.20), we have

$$\text{Var} \left(\sum_{j \in \mathcal{S}^c} \hat{A}_{ij} - A_{ij} \right) = N^2 O \left(\frac{h^4}{n h^{d_\zeta}} + h^2 \delta_{N,n}^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right) \quad (\text{C.21})$$

In addition, combining the bias we derived previously, (B.8), (B.15), (C.5), (C.6), (C.7), (C.13), (C.14), (C.15), and (C.19), we can find constants $\bar{E}_1 > 0$ and $c > 0$ such that

$$\mathbb{P} \left(\left| \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) \right| \leq \bar{E}_1 N \left(h^2 + \frac{h^2}{\sqrt{n h^{d_\zeta}}} + h \delta_{N,n} + \frac{1}{\sqrt{n}} + \frac{1}{n h^{d_\zeta}} \right) \right) \geq 1 - c n^{-1/2}$$

When W is uniformly bounded, we can extend the analysis and obtain similar results. ■

D Supplementary Proofs

Lemma D.1 Let $\zeta_i \in \{\bar{\zeta}_1, \dots, \bar{\zeta}_G\}$ be i.i.d. random variables such that $0 \leq \bar{\zeta}_g \leq 1$ for any $g = 1, 2, \dots, G$, then $f(\cdot, \cdot)$ satisfies Assumption 2 except in the degenerate case where there exist $g \neq g'$ such that $f(\bar{\zeta}_g, \zeta) = f(\bar{\zeta}_{g'}, \zeta)$ for all $\zeta \in \{\bar{\zeta}_1, \dots, \bar{\zeta}_G\}$.

Proof of Lemma D.1. Without loss of generality, suppose that $f(\bar{\zeta}_1, \zeta_{g'}) > 0$ for all g' , then for any $g, g' \in \{1, \dots, G\}$,

$$\begin{aligned} & \max_{\zeta \in \{\bar{\zeta}_1, \dots, \bar{\zeta}_G\}} \left| \int f(\zeta, \zeta') (f(\bar{\zeta}_g, \zeta') - f(\bar{\zeta}_{g'}, \zeta')) d\mathbb{P}(\zeta') \right| \\ & \geq \max \left\{ \left| \int f(\bar{\zeta}_g, \zeta') (f(\bar{\zeta}_g, \zeta') - f(\bar{\zeta}_{g'}, \zeta')) d\mathbb{P}(\zeta') \right|, \left| \int f(\bar{\zeta}_{g'}, \zeta') (f(\bar{\zeta}_g, \zeta') - f(\bar{\zeta}_{g'}, \zeta')) d\mathbb{P}(\zeta') \right| \right\} \end{aligned}$$

Then expression above equals 0 if and only if

$$\int f^2(\bar{\zeta}_g, \zeta) d\mathbb{P}(\zeta) = \int f^2(\bar{\zeta}_{g'}, \zeta) d\mathbb{P}(\zeta) = \int f(\bar{\zeta}_g, \zeta) f(\bar{\zeta}_{g'}, \zeta) d\mathbb{P}(\zeta)$$

which implies that $f(\bar{\zeta}_g, \zeta) = f(\bar{\zeta}_{g'}, \zeta)$ for all $\zeta \in \{\bar{\zeta}_1, \dots, \bar{\zeta}_G\}$. ■

Lemma D.2 Let $\zeta_i \in [0, 1]^{d_\zeta}$ be i.i.d. random vectors such that the components of $\zeta_i = (\zeta_{i1}, \dots, \zeta_{id_\zeta})'$ are mutually independent and each follows the uniform distribution on $[0, 1]$, then

$$f(\zeta_i, \zeta_j) = \frac{\exp(\omega(\zeta_i, \zeta_j)' \beta)}{1 + \exp(\omega(\zeta_i, \zeta_j)' \beta)}$$

satisfies Assumption 2 when

$$\omega(\zeta_i, \zeta_j)' \beta = \zeta_{i1} + \zeta_{j1} + \sum_{k=2}^d \beta_k |\zeta_{ik} - \zeta_{jk}| + \sum_{k=d+1}^{d_\zeta} \beta_k (\zeta_{ik} - \zeta_{jk})^2, \quad \beta_2, \dots, \beta_k \leq 0$$

Proof of Lemma D.2. Since $\zeta_i \in [0, 1]^{d_\zeta}$ be i.i.d. random vectors such that the components of $\zeta_i = (\zeta_{i1}, \dots, \zeta_{id_\zeta})'$ are mutually independent and each follows the uniform distribution on $[0, 1]$, it is sufficiently to check the Assumption 2 under the following three specific case.

Case 1: $d_\zeta = 1$ and $\omega(\zeta_i, \zeta_j) = \zeta_i + \zeta_j$ In this case, for each $\zeta_1 < \zeta_2 \in [0, 1]$ we have

$$f(\zeta_1, \zeta_2) = \frac{\exp(\zeta_1 + \zeta_2)}{1 + \exp(\zeta_1 + \zeta_2)}$$

and

$$\begin{aligned} d_{12} &= \sup_{\zeta} \left| \int f(\zeta, \zeta') (f(\zeta_1, \zeta') - f(\zeta_2, \zeta')) d\zeta' \right| \\ &\geq \left| \int f(0, \zeta') (f(\zeta_1, \zeta') - f(\zeta_2, \zeta')) d\zeta' \right| \\ &\geq \left(\inf_{\zeta \in [0, 1]} f(0, \zeta) \right) \left| \int_0^1 \int_{\zeta_1}^{\zeta_2} f'(\zeta, \zeta') d\zeta d\zeta' \right| \\ &\stackrel{(i)}{\geq} \left(\inf_{\zeta \in [0, 1]} f(0, \zeta) \right) \left(\inf_{\zeta, \zeta' \in [0, 1]} f'(\zeta, \zeta') \right) |\zeta_2 - \zeta_1| \end{aligned}$$

where inequality (i) holds because $f'(\cdot, \cdot) > 0$ on $[0, 1]^2$. It is straightforward to check that $(\inf_{\zeta \in [0, 1]} f(0, \zeta)) > 0$ and $(\inf_{\zeta, \zeta' \in [0, 1]} f'(\zeta, \zeta')) > 0$, thus, let $\lambda = (\inf_{\zeta \in [0, 1]} f(0, \zeta)) (\inf_{\zeta, \zeta' \in [0, 1]} f'(\zeta, \zeta'))$, we prove that

$$d_{12} \geq \lambda |\zeta_2 - \zeta_1|$$

Case 2: $d_\zeta = 1$ and $\omega(\zeta_i, \zeta_j) = -|\zeta_i - \zeta_j|$ In this case, for each $\zeta_1 < \zeta_2 \in [0, 1]$ we have

$$f(\zeta_1, \zeta_2) = \frac{\exp(-|\zeta_1 - \zeta_2|)}{1 + \exp(-|\zeta_1 - \zeta_2|)}$$

and define

$$h_0(\zeta) = \int f(0, \zeta') f(\zeta, \zeta') d\zeta', \quad h_1(\zeta) = \int f(1, \zeta') f(\zeta, \zeta') d\zeta'$$

and it follows that

$$d_{12} = \sup_{\zeta} \left| \int f(\zeta, \zeta') (f(\zeta_1, \zeta') - f(\zeta_2, \zeta')) d\zeta' \right| \geq \max \{ |h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)| \}$$

Thus, it suffices to prove that there exists a constant $\lambda > 0$ such that for any $\zeta_1, \zeta_2 \in [0, 1]$:

$$\max \{ |h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)| \} \geq \lambda |\zeta_1 - \zeta_2|$$

It is straightforward to check that

$$\begin{aligned} h'_0(\zeta) &= - \int f(0, \zeta') \text{sign}(\zeta - \zeta') f(\zeta, \zeta') (1 - f(\zeta, \zeta')) d\zeta' \\ h_0(\zeta) &= h_1(1 - \zeta), \quad h'_1(\zeta) = -h'_0(1 - \zeta) \end{aligned}$$

Also, $h_0(\cdot)$ achieves its maximum at $\zeta_0^* < 1/2$ and $h_1(\cdot)$ achieves its maximum at $\zeta_1^* = 1 - \zeta_0^* > 1/2$. $h_0(\cdot)$ is a strictly increasing function ζ when $\zeta < \zeta_0^*$, and is a strictly decreasing function ζ when $\zeta > \zeta_0^*$.

Step 1 $\zeta_1 \in [\zeta_0^*, \zeta_1^*] \vee \zeta_2 \in [\zeta_0^*, \zeta_1^*]$ Fix an arbitrary positive constant $\delta < 1/2 - \zeta_0^*$, define

$$\lambda_0 = \inf_{\zeta \in [\zeta_0^* + \delta, 1]} |h'_0(\zeta)|$$

One can check that $\lambda_0 > 0$ because $h'_0(\cdot)$ is strictly negative on $[\zeta_0^* + \delta, 1]$. Then:

- (i) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_0^*, \zeta_1^* - \delta_0]$, we have $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0 |\zeta_1 - \zeta_2|$ by mean value theorem.
- (ii) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_1^* - \delta_0, \zeta_1^*]$, $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0 (\zeta_1^* - \zeta_2^* - \delta)$ and thus, $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0 (\zeta_1^* - \zeta_2^* - \delta) |\zeta_1 - \zeta_2|$ because $|\zeta_1 - \zeta_2| \leq 1$.
- (iii) When $\zeta_1, \zeta_2 \in [\zeta_0^*, \zeta_1^*]$, since $\delta < 1/2 - \zeta_0^*$, it follows that

$$\max \{ |h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)| \} \geq \lambda_0 |\zeta_1 - \zeta_2|$$

- (iv) When $\zeta_1 \in [\zeta_0^*, \zeta_0^* + \delta]$ and $\zeta_2 \in [\zeta_1^*, 1]$, $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0 (\zeta_1^* - \zeta_2^* - \delta)$ and thus, $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0 (\zeta_1^* - \zeta_2^* - \delta) |\zeta_1 - \zeta_2|$ because $|\zeta_1 - \zeta_2| \leq 1$.
- (v) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_0^*, \zeta_1^* - \delta_0]$, we have $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0 |\zeta_1 - \zeta_2|$ by mean value theorem.

Therefore, we can show that

$$\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_0(\zeta_1^* - \zeta_0^* - \delta)|\zeta_1 - \zeta_2|$$

Step 2 $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$ Define

$$\lambda_1 = \inf_{\zeta \in [0, \zeta_0^*]} |h_0(\zeta) - h_1(\zeta)|$$

We have $\lambda_1 > 0$ because $h_0(\zeta) > h_1(\zeta)$ when $\zeta \in [0, \zeta_0^*]$. Thus, for any $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$, we have

(i) When $|h_0(\zeta_1) - h_0(\zeta_2)| \leq \lambda_1$, we have $|h_1(\zeta_1) - h_1(\zeta_2)| \geq 2\lambda_1 - \lambda_1 = \lambda_1$.

(ii) When $|h_1(\zeta_1) - h_1(\zeta_2)| \leq \lambda_1$, we have $|h_0(\zeta_1) - h_0(\zeta_2)| \geq 2\lambda_1 - \lambda_1 = \lambda_1$.

Therefore, we have $\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_1 \geq \lambda_1|\zeta_1 - \zeta_2|$ when $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$.

Combining the conclusion in two steps yields that

$$d_{12} \geq \max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \min\{\lambda_0(\zeta_1^* - \zeta_0^* - \delta), \lambda_1\}|\zeta_2 - \zeta_1|$$

Case 3: $d_\zeta = 1$ and $\omega(\zeta_i, \zeta_j) = -(\zeta_i - \zeta_j)^2$ In this case, for each $\zeta_1 < \zeta_2 \in [0, 1]$ we have

$$f(\zeta_1, \zeta_2) = \frac{\exp(-(\zeta_1 - \zeta_2)^2)}{1 + \exp(-(\zeta_1 - \zeta_2)^2)}$$

and define

$$h_0(\zeta) = \int f(0, \zeta')f(\zeta, \zeta')d\zeta', \quad h_1(\zeta) = \int f(1, \zeta')f(\zeta, \zeta')d\zeta'$$

and it follows that

$$d_{12} = \sup_{\zeta} \left| \int f(\zeta, \zeta')(f(\zeta_1, \zeta') - f(\zeta_2, \zeta'))d\zeta' \right| \geq \max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\}$$

Thus, it suffices to prove that there exists a constant $\lambda > 0$ such that for any $\zeta_1, \zeta_2 \in [0, 1]$:

$$\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda|\zeta_1 - \zeta_2|$$

It is straightforward to check that

$$\begin{aligned} h'_0(\zeta) &= -2 \int f(0, \zeta')(\zeta - \zeta')f(\zeta, \zeta')(1 - f(\zeta, \zeta'))d\zeta' \\ h_0(\zeta) &= h_1(1 - \zeta), \quad h'_1(\zeta) = -h'_0(1 - \zeta) \end{aligned}$$

Also, $h_0(\cdot)$ achieves its maximum at $\zeta_0^* < 1/2$ and $h_1(\cdot)$ achieves its maximum at $\zeta_1^* = 1 - \zeta_0^* > 1/2$. $h_0(\cdot)$ is a strictly increasing function ζ when $\zeta < \zeta_0^*$, and is a strictly decreasing function ζ when $\zeta > \zeta_0^*$.

Step 1 $\zeta_1 \in [\zeta_0^*, \zeta_1^*] \vee \zeta_2 \in [\zeta_0^*, \zeta_1^*]$ Fix an arbitrary positive constant $\delta < 1/2 - \zeta_0^*$, define

$$\lambda_0 = \inf_{\zeta \in [\zeta_0^* + \delta, 1]} |h'_0(\zeta)|$$

One can check that $\lambda_0 > 0$ because $h'_0(\cdot)$ is strictly negative on $[\zeta_0^* + \delta, 1]$. Then:

- (i) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_0^*, \zeta_1^* - \delta_0]$, we have $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0 |\zeta_1 - \zeta_2|$ by mean value theorem.
- (ii) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_1^* - \delta_0, \zeta_1^*]$, $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0(\zeta_1^* - \zeta_2^* - \delta)$ and thus, $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0(\zeta_1^* - \zeta_2^* - \delta) |\zeta_1 - \zeta_2|$ because $|\zeta_1 - \zeta_2| \leq 1$.
- (iii) When $\zeta_1, \zeta_2 \in [\zeta_0^*, \zeta_1^*]$, since $\delta < 1/2 - \zeta_0^*$, it follows that

$$\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_0 |\zeta_1 - \zeta_2|$$

- (iv) When $\zeta_1 \in [\zeta_0^*, \zeta_0^* + \delta]$ and $\zeta_2 \in [\zeta_1^*, 1]$, $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0(\zeta_1^* - \zeta_2^* - \delta)$ and thus, $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0(\zeta_1^* - \zeta_2^* - \delta) |\zeta_1 - \zeta_2|$ because $|\zeta_1 - \zeta_2| \leq 1$.
- (v) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_0^*, \zeta_1^* - \delta_0]$, we have $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0 |\zeta_1 - \zeta_2|$ by mean value theorem.

Therefore, we can show that

$$\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_0(\zeta_1^* - \zeta_0^* - \delta) |\zeta_1 - \zeta_2|$$

Step 2 $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$ Define

$$\lambda_1 = \inf_{\zeta \in [0, \zeta_0^*]} |h_0(\zeta) - h_1(\zeta)|$$

We have $\lambda_1 > 0$ because $h_0(\zeta) > h_1(\zeta)$ when $\zeta \in [0, \zeta_0^*]$. Thus, for any $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$, we have

- (i) When $|h_0(\zeta_1) - h_0(\zeta_2)| \leq \lambda_1$, we have $|h_1(\zeta_1) - h_1(\zeta_2)| \geq 2\lambda_1 - \lambda_1 = \lambda_1$.
- (ii) When $|h_1(\zeta_1) - h_1(\zeta_2)| \leq \lambda_1$, we have $|h_0(\zeta_1) - h_0(\zeta_2)| \geq 2\lambda_1 - \lambda_1 = \lambda_1$.

Therefore, we have $\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_1 \geq \lambda_1 |\zeta_1 - \zeta_2|$ when $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$.

Combining the conclusion in two steps yields that

$$d_{12} \geq \max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \min\{\lambda_0(\zeta_1^* - \zeta_0^* - \delta), \lambda_1\} |\zeta_2 - \zeta_1|$$

■

Lemma D.3 *There are two sets of independent sub-Gaussian random variables $\{X_i\}_{i=1}^N$ and $\{Y_i\}_{i=1}^N$, such that there exist $\sigma_X, \sigma_Y > 0$:*

$$\sup_{1 \leq i \leq N} \mathbb{E}(e^{\lambda X_i}) \leq e^{\frac{\lambda^2 \sigma_X^2}{2}}, \quad \sup_{1 \leq i \leq N} \mathbb{E}(e^{\lambda Y_i}) \leq e^{\frac{\lambda^2 \sigma_Y^2}{2}}, \quad \forall \lambda \in \mathbb{R}$$

$\{X_i\}_{i=1}^N$ and $\{Y_i\}_{i=1}^N$ are independent of each other. Then for all $\epsilon > 0$, we have:

$$\mathbb{P}\left(\left|\frac{1}{N} \sum_{i=1}^N \alpha_i X_i\right| \geq \epsilon\right) \leq 2 \exp\left(-\frac{N\epsilon^2}{2\sigma^2 \rho_\alpha^2 + \epsilon}\right)$$

In addition, suppose there is a set of independent random variables sub-Gaussian random variables $\{Y_i\}_{i=1}^N$ which is also independent of $\{X_i\}_{i=1}^N$ such that there exists $\sigma^2 > 0$:

$$\mathbb{P} \left(\left| \frac{1}{N} \sum_{i=1}^N X_i Y_i \right| \geq \epsilon \right) \leq 2 \exp \left(- \frac{N \epsilon^2}{2 (\sigma_X^2 \sigma_Y^2 + \sigma_X \sigma_Y \epsilon)} \right)$$

Proof. It is no harm to assume $\mathbb{E}(X_i) = \mathbb{E}(Y_i) = 0$ for all i . Then

$$\mathbb{E} \exp (\lambda X_i Y_i) = \mathbb{E} (\mathbb{E} (\exp (\lambda X_i Y_i) \mid X_i)) \leq \mathbb{E} \exp \left(\frac{\lambda^2 X_i^2 \sigma_Y^2}{2} \right) \leq \frac{1}{\sqrt{1 - \lambda^2 \sigma_Y^2 \sigma_X^2}}, \quad \forall 0 \leq \lambda^2 \leq \frac{1}{\sigma_Y^2 \sigma_X^2}$$

When $\lambda^2 < \frac{1}{\sigma_Y^2 \sigma_X^2}$, we have

$$\frac{1}{\sqrt{1 - \lambda^2 \sigma_X^2 \sigma_Y^2}} \leq \exp \left(\frac{\lambda^2 \sigma_X^2 \sigma_Y^2}{2(1 - \lambda^2 \sigma_X^2 \sigma_Y^2)} \right) \leq \exp \left(\frac{\lambda^2 \sigma_X^2 \sigma_Y^2}{2(1 - \lambda \sigma_X \sigma_Y)} \right)$$

Therefore, $X_i Y_i$ is satisfies Bernstein's condition because

$$\mathbb{E} (\lambda X_i Y_i) \leq \exp \left(\frac{\lambda^2 \sigma_X^2 \sigma_Y^2}{2(1 - \lambda \sigma_X \sigma_Y)} \right), \quad \forall |\lambda| < \frac{1}{\sigma_X \sigma_Y}$$

It immediately follows that

$$\mathbb{E} \left(\frac{1}{N} \sum_{i=1}^N \lambda X_i Y_i \right) \leq \exp \left(\frac{\lambda^2 \sigma_X^2 \sigma_Y^2}{2N(1 - \lambda \sigma_X \sigma_Y)} \right), \quad \forall |\lambda| < \frac{1}{\sigma_X \sigma_Y}$$

and

$$\mathbb{P} \left(\left| \frac{1}{N} \sum_{i=1}^N X_i Y_i \right| \geq \epsilon \right) \leq 2 \exp \left(- \frac{N \epsilon^2}{2 (\sigma_X^2 \sigma_Y^2 + \sigma_X \sigma_Y \epsilon)} \right)$$

■